

Amiodarone: Drug information - UpToDate

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For additional information see "Amiodarone: Patient drug information" and "Amiodarone: Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

ALERT: US Boxed Warning

Life-threatening arrhythmias (tablet):

Amiodarone is intended for use only in patients with indicated life-threatening arrhythmias because its use is accompanied by substantial toxicity.

Pulmonary toxicity (tablet):

Amiodarone can cause pulmonary toxicity (hypersensitivity pneumonitis or interstitial/alveolar pneumonitis) that has resulted in clinically manifest disease at rates as high as 17% in some series of patients. Pulmonary toxicity has been fatal about 10% of the time. Obtain a baseline chest X-ray and pulmonary function tests, including diffusion capacity, when amiodarone therapy is initiated. Repeat history, physical exam, and chest X-ray every 3 to 6 months.

Hepatotoxicity (tablet):

Amiodarone can cause hepatotoxicity, which can be fatal. Obtain baseline and periodic liver transaminases and discontinue or reduce dose if the increase exceeds 3 times normal or doubles in a patient with an elevated baseline. Discontinue amiodarone if the patient experiences signs or symptoms of clinical liver injury.

Worsened arrhythmia (tablet):

Amiodarone can exacerbate arrhythmias. Initiate amiodarone in a clinical setting where continuous ECGs and cardiac resuscitation are available.

Brand Names: US

- Nexterone;
- Pacerone

Brand Names: Canada

- APO-Amiodarone;
- DOM-Amiodarone [DSC];
- JAMP-Amiodarone;
- PMS-Amiodarone;

- PRO-Amiodarone-200;
- RIVA-Amiodarone;
- SANDOZ Amiodarone;
- TEVA-Amiodarone

Pharmacologic Category

- Antiarrhythmic Agent, Class III

Dosing: Adult

Supraventricular arrhythmias

Supraventricular arrhythmias:

Note: Amiodarone is a preferred antiarrhythmic for patients with structural heart disease.

Atrial fibrillation:

Pharmacologic cardioversion (off-label use) (alternative agent):

Note: Although amiodarone may slow ventricular response, the use of amiodarone (both IV and oral) is not particularly effective for cardioversion of atrial fibrillation to normal sinus rhythm as compared to other antiarrhythmics (eg, flecainide). Various IV and oral regimens have been described and may be used clinically. The more common regimens utilized include:

IV: 150 mg over 10 to 30 minutes or 300 mg over 60 minutes, then 1 mg/minute for 6 hours, then 0.5 mg/minute for 18 hours. Continue for a total loading dose of ~6 to 10 g; may finish load with oral dosing. Change to oral maintenance dose when clinically indicated (Ref).

Oral: 400 to 1,200 mg per day in 2 to 3 divided doses for a total loading dose of ~6 to 10 g (total of IV plus oral doses), then change to a maintenance dose (Ref). **Note:** May initiate loading regimen 2 to 6 weeks prior to direct current cardioversion to reduce the risk of atrial fibrillation recurrence (Ref). Also see "Switching to Oral Amiodarone After IV Administration."

Maintenance of sinus rhythm (off-label use): Oral: 100 to 200 mg once daily (Ref). Use the lower maintenance dose for older patients or patients with low body mass (Ref).

Prevention of postoperative atrial fibrillation and atrial flutter associated with cardiothoracic surgery (off-label use) (alternative agent):

Note: In patients who cannot receive a beta-blocker, amiodarone may be used. A variety of regimens have been used in clinical trials, including oral and IV regimens:

IV only:

Preoperative regimen: 150 mg once, followed by 0.4 mg/kg/hour (~0.5 mg/minute for a 70 kg patient) for 3 days prior to surgery and for 5 days postoperatively (Ref).

Postoperative regimen: Starting at postoperative recovery: 1 g infused over 24 hours for 2 days (Ref).

IV followed by Oral:

Postoperative regimen: IV: Starting 6 hours postoperatively: 1,050 mg IV loading dose over 24 hours, followed by 400 mg orally 3 times daily on postop days 1 through 4 (Ref).

Oral only: 200 mg 3 times daily for 7 days prior to surgery, followed by 200 mg once daily until hospital discharge (Ref).

Rate control (alternative agent) (off-label use):

Note: IV amiodarone may be considered for acute rate control in patients who are critically ill and/or in decompensated heart failure when beta blockers and nondihydropyridine calcium channel blockers are ineffective or contraindicated due to their hemodynamic effects (Ref).

IV: 150 or 300 mg over 1 hour followed by 10 to 50 mg/hour over 24 hours (Ref) **or** 150 mg over at least 10 to 30 minutes, then 1 mg/minute for 6 hours, then 0.5 mg/minute for 18 hours; administer

repeat boluses of 150 mg over 10 to 30 minutes as needed but no more than 8 additional boluses should be administered in any 24-hour period (Ref). **Note:** Mean daily doses >2.1 g/day have been associated with hypotension.

Oral: 400 to 1,200 mg per day in 2 to 3 divided doses for a total loading dose of ~6 to 10 g (total of IV plus oral doses), then change to a maintenance dose; usual maintenance dose: 100 to 200 mg once daily (Ref). Also see "Switching to Oral Amiodarone After IV Administration."

Supraventricular tachycardia (eg, atrioventricular nodal reentrant tachycardia, atrioventricular reentrant tachycardia, focal atrial tachycardia) (off-label use):

Note: Amiodarone is usually reserved for use when other therapies have failed or are contraindicated. In general, most patients do not require long-term treatment with antiarrhythmic therapy.

Pharmacologic cardioversion:

IV: 150 mg over 10 minutes, followed by 1 mg/minute for 6 hours, then 0.5 mg/minute for 18 hours or transition to oral dosing (Ref).

Oral : 400 to 600 mg per day in 2 to 3 divided doses for 2 to 4 weeks; in an inpatient monitored setting, initial oral doses up to 1.2 g daily in divided doses may be considered; administer a total loading dose of ~6 to 10 g (total of IV plus oral doses) then transition to a maintenance dose (Ref); usual maintenance dose: 100 to 200 mg once daily (Ref). Also see "Switching to Oral Amiodarone After IV Administration."

Ventricular arrhythmias

Ventricular arrhythmias:

Electrical storm and incessant ventricular tachycardia, hemodynamically stable (off-label use):

IV: 150 mg over 10 minutes (may repeat if necessary), followed by 1 mg/minute IV infusion for 6 hours, then 0.5 mg/minute for 18 additional hours or until switched to oral therapy. **Note:** Coadministration with a beta-blocker (eg, propranolol) is recommended (Ref).

Oral (following IV therapy): 400 to 1,200 mg per day in 2 to 3 divided doses for 1 to 2 weeks (loading dose of ~10 g [total of IV plus oral doses]), followed by 200 to 400 mg once daily; use the lowest effective dose possible for maintenance (Ref). Also see "Switching to Oral Amiodarone After IV Administration."

Nonsustained ventricular tachycardia, symptomatic (off-label use):

Note: Consider addition of a beta-blocker or nondihydropyridine calcium channel blocker (eg, diltiazem) before starting antiarrhythmic therapy (Ref).

Oral: 400 to 1,200 mg per day in 2 to 3 divided doses for 1 to 2 weeks (total loading dose of ~6 to 10 g), followed by 200 to 400 mg once daily; use the lowest effective dose possible for maintenance (Ref).

Prevention of implantable cardioverter defibrillator shocks (off-label use):

Note: Antiarrhythmic therapy for this indication has not been shown to reduce mortality, but may improve quality of life by reducing frequency of implantable cardioverter defibrillator (ICD) shocks. Amiodarone is the preferred antiarrhythmic for this use (Ref).

Oral: 400 mg twice daily for 2 weeks, followed by 400 mg once daily for 4 weeks, then 200 mg once daily in combination with a beta-blocker (Ref) **or** 400 to 1,200 mg per day in 2 to 3 divided doses for a total loading dose of ~6 to 10 g, then 200 to 400 mg once daily; use the lowest effective dose possible for maintenance (Ref).

Primary prevention of sudden cardiac death due to ventricular arrhythmias (off-label use):

Note: Consider addition of a beta-blocker before starting antiarrhythmic therapy. For patients eligible for an ICD with left ventricular dysfunction, but who cannot or refuse to have ICD implantation, amiodarone can be utilized (Ref).

Oral: 400 to 1,200 mg per day in 2 to 3 divided doses for 1 to 2 weeks for a total loading dose of ~6 to 10 g, then a maintenance dose of 200 to 400 mg once daily; use the lowest effective dose possible for maintenance (Ref).

Secondary prevention of sudden cardiac death due to ventricular arrhythmias (eg, ventricular fibrillation or hemodynamically unstable ventricular tachycardia):

Note: ICD implantation is preferred over antiarrhythmic therapy. In patients who do not meet ICD implant criteria, have contraindications, or refuse ICD implantation, antiarrhythmics can be utilized. Beta-blockers are recommended prior to or at the time of antiarrhythmic therapy initiation (Ref).

Oral: 400 to 1,200 mg per day in 2 to 3 divided doses for 1 to 2 weeks (loading dose of ~6 to 10 g [total of IV plus oral doses]), followed by a maintenance dose of 200 to 400 mg once daily; use the lowest effective dose possible for maintenance (Ref). Also see "Switching to Oral Amiodarone After IV Administration."

Sudden cardiac arrest due to ventricular fibrillation or pulseless ventricular tachycardia (unresponsive to CPR, defibrillation, and epinephrine):

IV push, Intraosseous (IO): Initial: 300 mg (undiluted) rapid bolus; if ventricular fibrillation or pulseless ventricular tachycardia continues after subsequent defibrillation attempt or reoccurs after initially achieving return of spontaneous circulation, administer supplemental dose of 150 mg (Ref).

Upon return of spontaneous circulation (ROSC): Continuous IV infusion: 1 mg/minute for 6 hours, then 0.5 mg/minute for 18 hours or until switched to oral therapy for **secondary prevention of sudden cardiac death due to ventricular arrhythmias** (Ref).

Sustained monomorphic ventricular tachycardia, hemodynamically stable (off-label use):

IV: 150 mg over 10 minutes, then 1 mg/minute for 6 hours, followed by 0.5 mg/minute for at least 18 hours **or** until switched to an oral maintenance regimen (see "Switching to Oral Amiodarone After IV Administration").

Breakthrough hemodynamically stable ventricular tachycardia: **IV:** 150 mg over 10 minutes (mean daily doses >2.1 g/day have been associated with hypotension).

Oral (following IV therapy): 400 to 1,200 mg per day in 2 to 3 divided doses for 1 to 2 weeks (loading dose of ~6 to 10 g [total of IV plus oral doses]), followed by a maintenance dose of 200 to 400 mg once daily (Ref); if arrhythmia remains controlled on 200 mg once daily, consider reducing dose further to 100 mg once daily to reduce the risk of adverse effects (Ref). Also see "Switching to Oral Amiodarone After IV Administration."

Ventricular premature beats, symptomatic (off-label use):

Note: For patients who are refractory to beta-blocker or nondihydropyridine calcium channel blocker therapy (eg, diltiazem) and cannot take class Ic antiarrhythmic agents (Ref).

Oral: 400 to 1,200 mg per day in 2 to 3 divided doses for 1 to 2 weeks (total loading dose of ~6 to 10 g), then when adequate control is achieved, decrease to 200 to 400 mg once daily; use lowest effective dose to minimize adverse effects (Ref).

Switching to oral amiodarone after IV administration: Use the following as a guide:

≤2-week IV infusion: 400 to 1,200 mg per day in 2 to 3 divided doses until loading dose of ~6 to 10 g (total of IV plus oral doses) has been administered, then start maintenance dose of 100 to 400 mg once daily (depending on indication) (Ref).

>2-week IV infusion: 100 to 400 mg once daily (depending on indication) (Ref).

Note: Conversion from IV to oral therapy has not been formally evaluated. Some experts recommend a 1- to 2-day overlap when converting from IV to oral therapy (Ref).

Switching to IV amiodarone after oral administration: Due to the long half-life, replacement therapy may not be necessary when oral maintenance doses are temporarily discontinued for only a few days. If oral maintenance doses must be withheld for ≥5 to 7 days, may consider administering

~50% (range of 30% to 70%) of the oral daily dose IV (based on bioavailability) as a slow bolus infusion once daily (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

Altered kidney function: Mild to severe impairment: No dosage adjustment necessary (Ref).

Hemodialysis, intermittent (thrice weekly): Poorly dialyzed (amiodarone and N-desethylamiodarone): No supplemental dose or dosage adjustment necessary (Ref).

Peritoneal dialysis: Poorly dialyzed (amiodarone and N-desethylamiodarone): No dosage adjustment necessary (Ref).

CRRT: No dosage adjustment necessary (Ref).

PIRRT (eg, sustained, low-efficiency diafiltration): No dosage adjustment necessary (Ref).

Dosing: Liver Impairment: Adult

The liver dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Matt Harris, PharmD, MHS, BCPS, FAST, FCCP; Jeong Park, PharmD, MS, BCTXP, FCCP, FAST; Arun Jesudian, MD; Sasan Sakiani, MD.

Note: Amiodarone is primarily metabolized in the liver with biliary excretion. Both the parent drug and its active metabolite (N-desethylamiodarone) are extensively distributed into tissues and demonstrate a prolonged elimination half-life (Ref). In cirrhosis, amiodarone may accumulate in the liver and amiodarone-induced liver injury (eg, secondary phospholipidosis) may persist after discontinuation (Ref). The IV formulation has been associated with acute hepatotoxicity, while chronic toxicity has been reported with prolonged oral use (Ref). In general, the oral route is preferred (Ref).

Liver impairment prior to treatment initiation:

Initial or dose adjustment in patients with preexisting liver cirrhosis:

Child-Turcotte-Pugh class A to C: IV, Oral: In general, other agents are preferred (Ref). If use is deemed necessary, no initial dosage adjustment is necessary. Use lowest effective dose for the shortest duration possible to mitigate the risk of liver accumulation and toxicity in collaboration with clinical experts (eg, cardiology, hepatology) (Ref).

Liver impairment developing in patients already receiving amiodarone:

Chronic disease progression (eg, outpatient):

Baseline to Child-Turcotte-Pugh class A to C: Although the manufacturer's labeling states to discontinue or reduce dose if liver transaminases exceed 3 times normal or double in a patient with elevated baseline transaminases, permanent discontinuation of amiodarone therapy as soon as clinically feasible in collaboration with appropriate clinical experts (eg, cardiology, hepatology) is preferred regardless of the increase in transaminase levels (Ref)

Amiodarone-induced liver injury: Permanently discontinue use if jaundice is present and/or AST or ALT increases to >3 times baseline and laboratory abnormalities cannot be explained by other etiologies (eg, alcohol, active viral hepatitis) (Ref). Amiodarone-induced liver injury may persist even after therapy has been discontinued (Ref).

Dosing: Older Adult

Refer to adult dosing. Avoid as first-line therapy for atrial fibrillation unless the patient has heart failure or substantial left ventricular hypertrophy (Ref). Dose selection should be cautious, at low end of dosage range, and titration should be slower to evaluate response. Although not supported

by clinical evidence, a maintenance dose of 100 mg daily is commonly used, especially for elderly patients or patients with low body mass (Ref).

Dosing: Pediatric

(For additional information see "Amiodarone: Pediatric drug information")

Perfusing wide-complex tachycardias

Perfusing wide-complex tachycardias: Note: Expert consultation recommended prior to use (Ref).

Infants, Children, and Adolescents: IV, Intraosseous: Loading dose: 5 mg/kg (maximum adult dose: 300 mg/dose) over 20 to 60 minutes (Ref).

Tachyarrhythmia, treatment

Tachyarrhythmia, treatment (acute, life-threatening): Limited data available:

Oral:

Infants: Oral: Loading dose: 10 to 20 mg/kg/day in two divided doses for 5 to 14 days; dosage should then be reduced to 5 to 7 mg/kg/day once daily (Ref).

Note: For infants, BSA-directed dosing has been suggested: Loading dose: 600 to 800 mg/**1.73 m²/day** in 1 to 2 divided doses (equivalent to 347 to 462 mg/m²/day); maintenance dose: 200 to 400 mg/**1.73 m²/day** once daily (equivalent to 116 to 231 mg/m²/day) (Ref).

Children and Adolescents: Oral: Loading dose: 10 mg/kg/day in two divided doses for 5 to 14 days; dosage should then be reduced to 5 to 7 mg/kg/day once daily (Ref).

IV:

Infants, Children, and Adolescents:

Loading dose: IV: 5 mg/kg (maximum dose: 300 mg/dose) given over 60 minutes; **Note:** Most studies used bolus infusion time of 60 minutes to avoid hypotension; may repeat initial loading dose to a maximum total initial load: 10 mg/kg; do not exceed total daily bolus of 15 mg/kg/**day** (Ref).

Note: Dividing the 5 mg/kg loading dose into 1 mg/kg aliquots (each administered over 5 to 10 minutes) has been used; an additional 1 to 5 mg/kg loading dose was given in the same manner, if needed, after 30 minutes (Ref).

Continuous IV infusion (if needed): **Note:** Reported dosing units for regimens are variable (**mcg/kg/minute** and **mg/kg/day**); ASHP recommends standard dosing units of **mcg/kg/minute**; use caution to ensure appropriate dose and dosing units are used; taper infusion as soon as clinically possible and switch to oral therapy if necessary.

Dosing based on **mcg/kg/minute**: Initial: 5 **mcg/kg/minute**; increase incrementally as clinically needed; range: 5 to 15 **mcg/kg/minute**; maximum daily dose: 2,200 mg/**day** (Ref).

Dosing based on **mg/kg/day**: Initial: 10 **mg/kg/day**; increase incrementally as clinically needed; range: 10 to 20 **mg/kg/day**; maximum daily dose: 2,200 mg/**day** (Ref).

Ventricular fibrillation or pulseless ventricular tachycardia, shock refractory

Ventricular fibrillation (VF) or pulseless ventricular tachycardia (VT), shock refractory:

Infants, Children, and Adolescents: IV, Intraosseous: 5 mg/kg (maximum dose: 300 mg/dose) rapid bolus; may repeat 5 mg/kg/dose (maximum dose: 150 mg/dose) twice for a total of 3 doses (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

Infants, Children, and Adolescents:

Altered kidney function: There are no dosage adjustments provided in the manufacturer's labeling; no dosage adjustment would be necessary (Ref).

Hemodialysis: Not dialyzable; supplemental dose is not necessary (Ref).

Peritoneal dialysis: Not dialyzable; supplemental dose is not necessary (Ref).

Dosing: Liver Impairment: Pediatric

Baseline: There are no dosage adjustments provided in the manufacturer's labeling. In adults with substantial liver impairment, dosage adjustment may be recommended (no guidelines exist).

Liver impairment during therapy: There are no dosage adjustments provided in the manufacturer's labeling; in adult patients, dosage adjustment is recommended.

Adverse Reactions (Significant): Considerations

Bradycardia/hypotension

Acute hemodynamic changes may occur with amiodarone (**bradycardia** and **hypotension**), particularly with IV administration (Ref). A variety of longer-term changes in conduction/heart rate may occur with amiodarone use, most commonly bradycardia, even in patients receiving low-dose chronic oral therapy. Significant disruptions in conduction, including cases of **atrioventricular block**, as well as **sinoatrial arrest** have been reported in adult and pediatric patients (Ref). Enhanced accessory pathway conduction has been reported (Ref). Bradycardia, kidney failure, atrioventricular nodal blockade, shock, and hyperkalemia (BRASH) syndrome has also been reported with amiodarone use (Ref).

Mechanism: Dose-related; beta-blocking and calcium channel blocking activity are responsible for bradycardia and sinoatrial blockade, as well as some of the hypotensive effects (Ref). Formulation-related; acute hypotension with rapid IV administration has been attributed in part to polysorbate 80 (Ref).

Onset: Acute hypotension during IV administration: Rapid; generally within 90 minutes but may occur up to 18 hours after initiation (Ref). Other hypotensive adverse reactions: Varied; generally reported in the first 12 months of therapy, while bradycardia may develop over months to years (Ref).

Risk factors:

Acute hypotension:

- Age >65 years with history of myocardial infarction (Ref)
- Pediatric patients: Age <3 months; baseline blood pressure below the third percentile for age and height; rapid IV bolus administration (ie, ≤20 minutes) (Ref)
- Formulations containing polysorbate 80 (Ref)

Hepatotoxicity

Hepatotoxicity has been reported with amiodarone and may cause two distinctly different toxicities, acute and chronic. Most cases of **acute hepatotoxicity** have occurred rapidly with IV administration (Ref). A case of acute hepatic failure following chronic oral use has been reported (Ref). Most cases of chronic toxicity have occurred with prolonged oral therapy (Ref). Hepatotoxicity may not be reversible (Ref). Fatal acute-on-chronic liver failure related to amiodarone-associated chronic steatohepatitis and microabscesses has also been reported (Ref).

Mechanism:

Acute toxicity: Not clearly established; multiple proposed mechanisms, including idiosyncratic reaction, hypersensitivity, free-radical formation, and direct toxicity of the vehicle for select IV formulations (ie, polysorbate 80) (Ref).

Chronic toxicity: Not clearly established; may be related to direct damage to lipid layers and lysosomal and/or mitochondrial dysfunction (Ref).

Onset:

Acute toxicity: Rapid; most cases have occurred within 24 hours of initiation following IV administration (Ref).

Chronic toxicity: Delayed; most cases occur after at least a year of oral therapy (Ref).

Risk factors:

Acute toxicity:

- Polysorbate 80 diluent in select IV formulations (Ref)
- Hepatic congestion (Ref)
- History of alcohol use disorder (potential risk factor, presumably because of depleted glutathione stores) (Ref)

Chronic toxicity:

- Cumulative dose (potential risk factor) (Ref)

Proarrhythmic effects

Amiodarone may exacerbate preexisting arrhythmia; may also cause new-onset **ventricular tachycardia**, life-threatening polymorphic ventricular tachycardia associated with **prolonged QT interval on ECG** and **torsades de pointes** (TdP), and **ventricular fibrillation** (Ref). Amiodarone may also increase resistance to cardioversion (Dorian 2003, Zhou 1998). Overall, proarrhythmic effects are low due to fast phase III repolarization, a low incidence on dispersion of repolarization, a lower potential to induce early after depolarization, and a weak effect on reverse frequency dependence (Ref).

Mechanism: Related to the pharmacologic action; type III antiarrhythmic that blocks potassium channels, resulting in the prolongation of myocardial repolarization, represented by prolongation of the QT interval (Ref).

Onset: Varied; generally, proarrhythmic events occur sporadically later in therapy (Ref). Rare cases have been reported after short-term, low-dose therapy (200 mg/day for 4 to 5 days) (Ref).

Risk factors:

Drug-induced QTc prolongation/TdP (in general):

- Females (Ref)
- Age >65 years (Ref)
- Structural heart disease (eg, history of myocardial infarction or heart failure with reduced ejection fraction) (Ref)
- Genetic defects of cardiac ion channels (Ref)
- History of drug-induced TdP (Ref)
- Congenital long QT syndrome (LQTS) (Ref)
- Baseline QTc interval prolongation (eg, >500 msec) or lengthening of the QTc by ≥ 60 msec (Ref)
- Electrolyte disturbances (eg, hypocalcemia, hypokalemia, or hypomagnesemia) (Ref)
- Bradycardia (Ref)
- Hepatic impairment (Ref)
- Kidney impairment (Ref)
- Loop diuretic use (Ref)
- Sepsis (Ref)

- Concurrent administration of multiple medications (≥ 2) that prolong the QT interval or medications with drug interactions that increase serum concentrations of QT-prolonging medications (Ref)

Pulmonary toxicity

Pulmonary toxicity may occur with amiodarone use and include a variety of clinical syndromes which may be categorized by onset. Acute events include pneumonitis, **pulmonary alveolar hemorrhage**, and **acute respiratory distress syndrome** (ARDS) (Ref). Pulmonary adverse reactions associated with longer-term therapy (subacute or chronic) include chronic eosinophilic pneumonia, chronic organizing pneumonia, acute fibrinous organizing pneumonia, and **pulmonary fibrosis** (Ref).

Mechanism:

Acute toxicity: Non-dose-related (idiosyncratic); may be related to production of superoxide radicals and release of active iodide species (Ref).

Chronic toxicity: May be mediated by a combination of direct toxicity to specific pulmonary cells, immune activation, and activation of angiotensin (Ref).

Onset:

Acute toxicity: Varied; may occur as early as 2 days to 2 weeks (Ref).

Chronic toxicity: Delayed; pulmonary fibrosis is usually identified after months to years of therapy (Ref).

Risk factors:

Acute toxicity:

- Cardiothoracic surgery (Ref)
- Exposure to a high inspiratory fraction of inhaled oxygen (FiO_2) (Ref)
- Pulmonary angiography (Ref)

Chronic toxicity:

- Higher daily and cumulative doses (Ref)
- Chronic kidney disease (Ref)
- Older adults (particularly >60 years of age) (Ref)

Thyroid effects

Amiodarone is associated with complex effects on thyroid function, resulting in either **hyperthyroidism** or **hypothyroidism**. Most euthyroid patients exhibit transient abnormalities of thyroid tests without overt symptoms and return to a baseline in 3 to 6 months (Ref). In a subset of patients, often in those with underlying thyroid disease, these effects may result in clinical thyroid disease, including rare cases of **thyrotoxicosis**, thyroid storm, or **myxedema** (including myxedema coma) (Ref).

Mechanism:

Hyperthyroidism: Two types of amiodarone-induced thyrotoxicosis are generally recognized:

- Type 1: Occurs in patients with previously known thyroid disease (Graves' disease, multinodular goiter); increased iodine intake increases synthesis and release of thyroid hormones (Ref).
- Type 2: Occurs in patients with previously healthy thyroid; amiodarone-mediated destructive thyroiditis increases release of previously synthesized thyroid hormones (Ref).

Hypothyroidism : Multiple mechanisms may contribute; thyroid synthetic activity inhibition, inability to escape the Wolff-Chaikoff effect, inhibition of deiodination (decreasing T_4 to T_3 conversion), inhibition of T_3 binding to its nuclear receptor, and a non-specific effect of iodine and amiodarone on thyroid follicles (Ref).

Onset: Delayed; most cases of hyperthyroidism occur between 4 months and 3 years of therapy and most cases of hypothyroidism occur within the first 6 to 12 months of therapy (Ref). In a pediatric series, the majority of thyroid dysfunction was noted within 35 days of amiodarone initiation (Ref).

Risk factors:

Hyperthyroidism:

- Iodine deficient region (Ref)
- Males (Ref)
- Thyroid disease (eg, Graves' disease, multinodular goiter) (Ref)
- Dilated cardiomyopathy (Ref)
- Cardiac sarcoidosis (Ref)

Hypothyroidism:

- Iodine sufficient region (Ref)
- Females (Ref)
- Hashimoto thyroiditis (Ref)
- Thyroid autoantibodies (Ref)

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified. Adverse reactions reported for the oral dosage form unless otherwise specified. Reported adverse reactions are for adults.

>10%:

Cardiovascular: Hypotension (refractory in rare cases)

Gastrointestinal: Nausea, vomiting (oral: >10%; IV: <2%)

Ophthalmic: Epithelial keratopathy (98% to 99%; vortex) (Ref)

Respiratory: Pulmonary toxicity (including hypersensitivity pneumonitis, interstitial/alveolar pneumonitis: ≤17%)

1% to 10%:

Cardiovascular: Bradycardia (2% to 4%), cardiac arrhythmia, edema, exacerbation of cardiac arrhythmia (2% to 5%), flushing, heart failure (IV, oral), prolonged QT interval on ECG (associated with worsening arrhythmia), shock (IV: <2%), sinus node dysfunction, torsades de pointes (IV: <2%), ventricular fibrillation (IV: <2%)

Dermatologic: Skin photosensitivity, solar dermatitis, Stevens-Johnson syndrome (IV, oral: <2%) (Ref)

Endocrine & metabolic: Decreased libido, hyperthyroidism (2%), hypothyroidism

Gastrointestinal: Abdominal pain, altered salivation, anorexia, constipation, diarrhea (IV: <2%), dysgeusia

Hepatic: Abnormal hepatic function tests (IV, oral)

Nervous system: Abnormal gait, altered sense of smell, ataxia, dizziness, fatigue, headache, insomnia, involuntary body movements, malaise, paresthesia, sleep disturbance, tremor

Ophthalmic: Blurred vision, visual disturbance, visual halos around lights

Renal: Kidney impairment (IV: <2%)

Respiratory: Acute respiratory distress syndrome (IV, oral: ≤2%; incidence may be higher in patients following anesthesia with high FiO₂ exposure) (Ref), pulmonary fibrosis

<1%:

Dermatologic: Alopecia, blue-gray skin pigmentation (incidence may be higher ~8%) (Ref), skin rash, spontaneous ecchymoses

Hepatic: Hepatic necrosis (acute: IV)

Frequency not defined:

Cardiovascular: Asystole (IV), atrioventricular block (IV), cardiogenic shock (IV), ventricular tachycardia (IV, oral)

Hematologic & oncologic: Disorder of hemostatic components of blood

Postmarketing (oral and/or IV):

Cardiovascular: Cardiac conduction disorder (including bundle branch block, infra-HIS block, and antegrade conduction via an accessory pathway) (Ref), sinoatrial arrest, sinoatrial block, vasculitis (Ref)

Dermatologic: Bullous dermatitis, eczema, erythema multiforme (Ref), erythema of skin (palmer) (Ref), exfoliative dermatitis, malignant neoplasm of skin (Ref), toxic epidermal necrolysis (Ref), urticaria (Ref)

Endocrine & metabolic: Malignant neoplasm of thyroid (Ref), myxedema (including myxedema coma) (Ref), SIADH (Ref), thyroid nodule, thyroid storm (Ref), thyrotoxicosis (Ref)

Gastrointestinal: Cholestasis (Ref), pancreatitis (including acute pancreatitis) (Ref), xerostomia

Genitourinary: Epididymitis (Ref), erectile dysfunction, orchitis (Ref)

Hematologic & oncologic: Agranulocytosis (Ref), aplastic anemia (Ref), granulocytosis, hemolytic anemia, immune thrombocytopenia (Ref), neutropenia (Ref), pancytopenia (Ref), thrombocytopenia

Hepatic: Acute hepatic failure (Ref), acute hepatotoxicity (Ref), cholestatic hepatitis (Ref), hepatic cirrhosis (Ref), hepatotoxicity (idiosyncratic) (Ref)

Hypersensitivity: Anaphylactic shock (Ref), anaphylaxis (Ref), angioedema (Ref), drug reaction with eosinophilia and systemic symptoms, nonimmune anaphylaxis

Local: Infusion-site reaction (including cellulitis, edema, erythema, granuloma, hypoesthesia, induration, inflammation, pain, phlebitis [IV concentrations of >3 mg/mL have been associated with peripheral vein phlebitis], pigment changes, pruritus, skin sloughing, thrombophlebitis, thrombosis, urticaria) (Ref)

Nervous system: Delirium (Ref), demyelinating disease (polyneuropathy) (Ref), hallucination, hypoesthesia, increased intracranial pressure, intracranial hypertension (Ref), myasthenia, parkinsonism (Ref), peripheral neuropathy (chronic treatment, partially to completely reversible with discontinuation) (Ref)

Neuromuscular & skeletal: Back pain, lupus-like syndrome (Ref), muscle spasm, myopathy, rhabdomyolysis

Ophthalmic: Corneal deposits, dry eye syndrome (Ref), optic neuritis, optic neuropathy (Ref)

Renal: Acute kidney injury (Ref)

Respiratory: Bronchospasm (Ref), cryptogenic organizing pneumonia, hemoptysis (Ref), hypoxia, pleural effusion (Ref), pleurisy, pulmonary alveolar hemorrhage (Ref), pulmonary infiltrates

Miscellaneous: Drug-induced phospholipidosis, fever, granuloma (of bone) (Ref), mass (pulmonary) (Ref)

Contraindications

Hypersensitivity to amiodarone, iodine, or any component of the formulation; sick sinus syndrome, second- or third-degree atrioventricular block, bradycardia leading to syncope without a functioning pacemaker; cardiogenic shock; preexcited atrial fibrillation/flutter (ACC/AHA [Joglar 2024]).

Note: The FDA-approved product labeling states amiodarone is contraindicated in patients with iodine hypersensitivity. This does not include most patients with allergic reactions to shellfish or contrast media, which are usually not due to iodine itself (Beall 2007; Brouse 2005; Lakshmanadoss 2012). However, exercise caution in patients with severe allergies to shellfish or contrast media (Brouse 2005).

Canadian labeling (oral formulation): Additional contraindications (not in US labeling): Evidence of hepatitis; pulmonary interstitial abnormalities; thyroid dysfunction.

Warnings/Precautions

Concerns related to adverse effects:

- Extravasation: May be a vesicant; ensure proper needle or catheter placement prior to infusion. Avoid extravasation.
- Ocular effects: Regular ophthalmic examination (including slit lamp and fundoscopy) is recommended. May cause optic neuropathy and/or optic neuritis resulting in visual impairment (peripheral vision loss, changes in acuity) at any time during therapy; permanent blindness has occurred. If symptoms of optic neuropathy and/or optic neuritis occur, prompt ophthalmic evaluation is recommended. If diagnosis of optic neuropathy and/or optic neuritis is confirmed, reevaluate amiodarone therapy. Corneal microdeposits occur in a majority of adults and may cause visual disturbances in up to 10% of patients (blurred vision, halos); asymptomatic microdeposits may be reversible and are not generally considered a reason to discontinue treatment. Corneal refractive laser surgery is generally contraindicated in amiodarone users (from manufacturers of surgical devices).
- Photosensitivity: Avoid excessive exposure to sunlight; may cause photosensitivity. During long-term treatment, a blue-gray discoloration of exposed skin may occur; risk increased in patients with fair complexion or excessive sun exposure; may be related to cumulative dose and duration of therapy.

Disease-related concerns:

- Arrhythmias: Appropriate use: The 2025 ACLS guidelines recommend the consideration of IV amiodarone as the preferred antiarrhythmic for the treatment of pulseless ventricular tachycardia/ventricular fibrillation unresponsive to CPR, defibrillation, and vasopressor therapy (AHA [Wigginton 2025]). In patients with non-life-threatening arrhythmias (eg, atrial fibrillation), amiodarone should be used only if the use of other antiarrhythmics has proven ineffective or are contraindicated.
- Cardiac devices (eg, implanted defibrillators, pacemakers): Chronic administration of antiarrhythmic drugs may affect defibrillation or pacing thresholds. Assess when initiating amiodarone and during therapy.
- Electrolyte imbalance: Correct electrolyte disturbances, especially hypokalemia, hypomagnesemia, or hypocalcemia, prior to use and throughout therapy.
- Myocardial infarction: In the setting of acute myocardial infarction (MI), beta-blocker therapy should be initiated even though concomitant amiodarone therapy provides beta-blockade. In the Cardiac Arrhythmia Suppression Trial (CAST), post-MI patients with asymptomatic, nonlife-threatening ventricular arrhythmias did not benefit and may have been harmed by attempts to suppress arrhythmia with flecainide or encainide. Although use of amiodarone post-MI was not associated with an increase in mortality in 2 trials, antiarrhythmic agents should be reserved for patients with life-threatening ventricular arrhythmias.

Special populations:

- Heart transplant patients: May cause an increased risk of primary graft dysfunction after heart transplantation in patients who received amiodarone prior to the transplant; occurred within 24 hours after transplant with no known secondary cause. Severe primary graft dysfunction may be irreversible. Consider alternate antiarrhythmic therapy in patients on the heart transplant waiting list.

- Surgical patients: Use caution and close perioperative monitoring in surgical patients; may enhance myocardial depressant and conduction effects of halogenated inhalational anesthetics; adult respiratory distress syndrome (ARDS) has been reported postoperatively (fatal in rare cases). Hypotension upon discontinuation of cardiopulmonary bypass during open-heart surgery have been reported (rare); relationship to amiodarone is unknown.

Dosage form specific issues:

- Benzyl alcohol and derivatives: Some dosage forms may contain benzyl alcohol; large amounts of benzyl alcohol (≥ 99 mg/kg/day) have been associated with a potentially fatal toxicity ("gasping syndrome") in neonates; the "gasping syndrome" consists of metabolic acidosis, respiratory distress, gasping respirations, CNS dysfunction (including convulsions, intracranial hemorrhage), hypotension, and cardiovascular collapse (AAP ["Inactive" 1997]; CDC 1982); some data suggest that benzoate displaces bilirubin from protein binding sites (Ahlfors 2001); avoid or use dosage forms containing benzyl alcohol with caution in neonates. See manufacturer's labeling.
- Commercially prepared premixed infusion: Contains the excipient cyclodextrin (sulfobutyl ether beta-cyclodextrin), which may accumulate in patients with renal insufficiency, although the clinical significance of this finding is uncertain (Luke 2010).
- Long-term use: There has been limited experience in patients receiving IV amiodarone for >3 weeks.
- Polysorbate 80: Some dosage forms may contain polysorbate 80 (also known as Tweens). Hypersensitivity reactions, usually a delayed reaction, have been reported following exposure to pharmaceutical products containing polysorbate 80 in certain individuals (Isaksson 2002; Lucente 2000; Shelley 1995). Thrombocytopenia, ascites, pulmonary deterioration, and renal and hepatic failure have been reported in premature neonates after receiving parenteral products containing polysorbate 80 (Alade 1986; CDC 1984). See manufacturer's labeling. Commercially prepared premixed amiodarone solutions do not contain polysorbate 80 and may have a lower incidence of hypotension.

Other warnings/precautions:

- Discontinuation of therapy: Patients may still be at risk for amiodarone-related adverse reactions or drug interactions after the drug has been discontinued. The pharmacokinetics are complex (due to prolonged duration of action and half-life) and difficult to predict.

Dosage Forms Considerations

Vials for injection contain benzyl alcohol which has been associated with "gasping syndrome" in neonates. Commercially-prepared premixed solutions do not contain benzyl alcohol.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Solution, Intravenous, as hydrochloride:

Nexterone: 150 mg/100 mL in Dextrose (100 mL); 360 mg/200 mL in Dextrose (200 mL)

Generic: 150 mg/3 mL (3 mL); 450 mg/9 mL (9 mL); 900 mg/18 mL (18 mL)

Tablet, Oral, as hydrochloride:

Pacerone: 100 mg

Pacerone: 200 mg [scored; contains fd&c red #40 (allura red ac dye), fd&c yellow #6 (sunset yellow)]

Pacerone: 400 mg [DSC] [scored; contains corn starch]

Generic: 100 mg, 200 mg, 400 mg

Generic Equivalent Available: US

Yes

Pricing: US

Solution (Amiodarone HCl Intravenous)

150 mg/3 mL (per mL): \$0.80 - \$1.76

450 mg/9 mL (per mL): \$0.73 - \$1.19

900 mg/18 mL (per mL): \$0.93 - \$1.15

Solution (Nexterone Intravenous)

150MG/100ML 4.21% (per mL): \$0.42

360MG/200ML 4.14% (per mL): \$0.28

Tablets (Amiodarone HCl Oral)

100 mg (per each): \$7.43

200 mg (per each): \$3.30 - \$3.48

400 mg (per each): \$1.60 - \$7.48

Tablets (Pacerone Oral)

100 mg (per each): \$7.43

200 mg (per each): \$7.27

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Intravenous:

Generic: 50 mg/mL (3 mL, 9 mL, 18 mL)

Solution, Intravenous, as hydrochloride:

Generic: 150 mg/3 mL (3 mL)

Tablet, Oral, as hydrochloride:

Generic: 100 mg, 200 mg

Extemporaneous Preparations

5 mg/mL Oral Suspension (ASHP Standard Concentration) (ASHP 2017)

A 5 mg/mL oral suspension may be made with tablets and either a 1:1 mixture of Ora-Sweet and Ora-Plus or a 1:1 mixture of Ora-Sweet SF and Ora-Plus adjusted to a pH between 6-7 using a sodium bicarbonate solution (5 g/100 mL of distilled water). Crush five 200 mg tablets in a mortar and reduce to a fine powder. Add small portions of the chosen vehicle and mix to a uniform paste; mix while adding the vehicle in incremental proportions to **almost** 200 mL; transfer to a calibrated bottle, rinse mortar with vehicle, and add quantity of vehicle sufficient to make 200 mL. Label "shake well" and "protect from light." Stable for 42 days at room temperature or 91 days refrigerated (preferred) (Ref).

Administration: Adult

Oral: Administer consistently with regard to meals. Take in divided doses with meals if GI upset occurs or if taking large daily dose ($\geq 1,000$ mg). If GI intolerance occurs with single-dose therapy, use twice daily dosing.

Enteral feeding tube:

The following recommendations are based upon the best available evidence and clinical expertise. Senior editorial team: Joseph I. Boullata, PharmD, RPh, CNS-S, FASPEN, FACN; Peggi A. Guenter, PhD, RN, FASPEN; Kathleen Gura, PharmD, BCNSP, FASHP, FASPEN, FPPA, FMSHP; Mark G. Klang, MS, RPh, BCNSP, PhD, FASPEN; Linda Lord, NP, ACNP-BC, CNSC, FASPEN; Lucas E. Orth, PharmD, BCPPS; Russel J. Roberts, PharmD, BCCCP, FCCM.

Oral tablet:

Note: Enteral feeding tube administration utilizing amiodarone tablets is not preferred; administration of crushed tablets has resulted in unpredictable drug absorption and decreased serum concentrations compared to oral administration (Ref). When alternatives are not available and use of tablets is deemed necessary for feeding tube administration, some institutions have successfully administered tablets; consider the risks versus benefits, maintain consistent administration practices (eg, timing and proximity to enteral nutrition [EN]), and ensure adequate flushing occurs following administration; monitor closely for therapeutic effect (Ref).

Gastric (eg, NG, G-tube) or post-pyloric (eg, J-tube) tubes: Crush tablet(s) into a fine powder and disperse in ≥ 15 mL purified water immediately prior to administration; draw up mixture into enteral dosing syringe and administer via feeding tube (Ref).

General guidance: Hold EN during amiodarone administration (Ref). Flush feeding tube with an appropriate volume of purified water (eg, 15 mL) before administration (Ref). Following administration, rinse container used for preparation with purified water; draw up rinse and administer contents to ensure delivery of entire dose (Ref). Flush feeding tube with an appropriate volume of purified water (eg, 15 mL) and restart EN (Ref).

Note: Recommendations may not account for differences in inactive ingredients, osmolality, or other formulation properties that may vary among products from different manufacturers.

IV: For infusions >1 hour, use concentrations ≤ 2 mg/mL unless a central venous catheter is used; commercially prepared premixed solutions in concentrations of 1.5 mg/mL and 1.8 mg/mL are available. Use only volumetric infusion pump; use of drop counting may lead to underdosage. Administer through an IV line located as centrally as possible. For peripheral infusions, an in-line filter has been recommended during administration to reduce the incidence of phlebitis (consult individual institutional policies and procedures). During pulseless VT/VF, administering **undiluted** is preferred (Ref). *The Handbook of Emergency Cardiovascular Care* (Ref) and the ACLS guidelines do not make any specific recommendations regarding dilution of amiodarone in this setting.

Adjust administration rate to urgency (give more slowly when perfusing arrhythmia present). Slow the infusion rate if hypotension or bradycardia develops. Infusions >2 hours must be administered in a non-PVC container (eg, glass or polyolefin). PVC tubing is recommended for administration regardless of infusion duration. **Incompatible** with heparin; flush with 0.9% sodium chloride prior to and following infusion. **Note:** IV administration at lower flow rates and higher concentrations than recommended may result in leaching of plasticizers (DEHP) from intravenous tubing. DEHP may adversely affect male reproductive tract development. Alternative means of dosing and administration (1 mg/kg aliquots) may need to be considered.

May be a vesicant; ensure proper needle or catheter placement prior to and during IV infusion. Avoid extravasation.

Extravasation management: If extravasation occurs, stop infusion immediately; leave needle/cannula in place temporarily but do **NOT** flush the line; gently aspirate extravasated solution, then remove needle/cannula; elevate extremity; apply dry warm compresses; initiate hyaluronidase antidote for refractory cases in addition to supportive management (Ref).

Hyaluronidase: Intradermal or SUBQ: Inject a total of 1 mL (15 units/mL) as 5 separate 0.2 mL injections (using a tuberculin syringe) around the site of extravasation; if IV catheter remains in place, administer IV through the infiltrated catheter; may repeat in 30 to 60 minutes if no resolution (Ref).

Preparation for Administration: Adult

Injection must be diluted in D5W before continuous IV infusion use. Dilute to final concentration of 1 to 6 mg/mL. Premixed solutions are available.

Usual Infusion Concentrations: Adult

Note: Premixed solutions available.

IV bolus or infusion: 1.5 mg/mL (eg, 150 mg in 100 mL of D5W) or 1.8 mg/mL (eg, 360 mg in 200 mL of D5W).

Administration: Pediatric

The following feeding tube recommendations are based upon the best available evidence and clinical expertise. Senior editorial team: Joseph I. Boullata, PharmD, RPh, CNS-S, FASPEN, FACN; Peggi A. Guenter, PhD, RN, FASPEN; Kathleen Gura, PharmD, BCNSP, FASHP, FASPEN, FPPA, FMSHP; Mark G. Klang, MS, RPh, BCNSP, PhD, FASPEN; Linda Lord, NP, ACNP-BC, CNSC, FASPEN; Lucas E. Orth, PharmD, BCPPS; Russel J. Roberts, PharmD, BCCCP, FCCM.

Note: Recommendations may not account for differences in inactive ingredients, osmolality, or other formulation properties that may vary among products from different manufacturers.

Oral:

Tablet: Administer at same time in relation to meals. In adults, may administer in divided doses with meals if GI upset occurs or if taking large daily dose.

Administration via feeding tube: Enteral feeding tube administration utilizing amiodarone tablets is not preferred; administration of crushed tablets has resulted in unpredictable drug absorption and decreased serum concentrations compared to oral administration (Ref). When alternatives are not available and use of tablets is deemed necessary for feeding tube administration, some institutions have successfully administered tablets; consider the risks versus benefits, maintain consistent administration practices (eg, timing and proximity to enteral nutrition), and ensure adequate flushing occurs following administration; monitor closely for therapeutic effect (Ref).

Gastric (eg, NG, G-tube) or post-pyloric (eg, J-tube) tubes: Crush tablet(s) into a fine powder and disperse in ≥ 15 mL purified water immediately prior to administration; draw up mixture into enteral dosing syringe and administer via feeding tube (Ref).

General guidance: Hold enteral nutrition during amiodarone administration (Ref). Flush feeding tube with the lowest volume of purified water necessary to clear the tube prior to administration based on size of patient and/or feeding tube (eg, neonates: 1 to 3 mL; infants and children: 2 to 5 mL; adolescents: 15 mL); refer to institutional policies and procedures (Ref). Following administration, rinse container used for preparation with purified water; draw up rinse and administer contents to ensure delivery of entire dose (Ref). Flush feeding tube with an appropriate volume of purified water and restart enteral nutrition (Ref).

IV: Adjust administration rate to patient's clinical condition and urgency; give slowly to patients who have a pulse (ie, perfusing arrhythmia). With perfusing arrhythmias (eg, atrial fibrillation, stable ventricular tachycardia), do not exceed recommended IV concentrations or rates of infusion listed below (severe hepatic toxicity may occur). Slow the infusion rate if hypotension or bradycardia develops. An in-line filter has been recommended during administration to reduce the incidence of phlebitis associated with dilution-induced precipitation (consult individual institutional policies and procedures) (Ref).

May be a vesicant; ensure proper needle or catheter placement prior to and during IV infusion. Avoid extravasation. If extravasation occurs, stop infusion immediately, leave needle/cannula in place temporarily but do **NOT** flush the line; gently aspirate extravasated solution, then remove needle/cannula

unless needed for IV antidote; elevate extremity; apply dry, warm compresses; initiate hyaluronidase antidote for refractory cases in addition to supportive management (Ref).

Pulseless VT or VF: Infants, Children, and Adolescents: Administer via rapid IV bolus; adult data in this setting suggest administration of undiluted drug may be preferred (Ref). **Note:** PALS guidelines (Ref) do not specify dilution of bolus dose and *The Handbook of Emergency Cardiovascular Care* (Ref) and the ACLS guidelines do not make any specific recommendations regarding dilution of amiodarone in this setting.

Perfusing arrhythmias: Information based on adult data: Injection must be diluted before IV use. Administer via central venous catheter, if possible; increased phlebitis may occur with peripheral infusions >3 mg/mL in D5W; the use of a central venous catheter with concentrations >2 mg/mL for infusions >1 hour is recommended. Must be infused via volumetric infusion device; drop size of IV solution may be reduced and underdosage may occur if drop counter-infusion sets are used. PVC tubing is recommended for administration regardless of infusion duration. **Incompatible** with heparin; flush with normal saline prior to and following infusion. **Note:** IV administration at lower flow rates (potentially associated with use in pediatrics) and higher concentrations than recommended may result in leaching of plasticizers (DEHP) from IV tubing. DEHP may adversely affect male reproductive tract development. Alternative means of dosing and administration (1 mg/kg aliquots) may need to be considered.

Neonates: Administer loading dose over 60 minutes; may be followed by continuous infusion (Ref).

Infants, Children, and Adolescents: Administer loading dose over 20 to 60 minutes (Ref); may be followed by continuous IV infusion.

Preparation for Administration: Pediatric

IV: **Perfusing arrhythmias:** Continuous IV infusion: Information based on adult data: Injection must be diluted in D5W to maximum concentration for IV infusion of 6 mg/mL prior to administration. Commercially prepared premixed solutions in concentrations of 1.5 mg/mL and 1.8 mg/mL are available. ASHP recommends standard concentrations of 1.8 mg/mL and 3.6 mg/mL in pediatric patients (Ref).

Note: Increased phlebitis may occur with peripheral infusions >3 mg/mL in D5W, and concentrations ≤2.5 mg/mL may be less irritating. Solutions that will infuse for >2 hours must be prepared in a non-PVC container (eg, glass or polyolefin).

Usual Infusion Concentrations: Pediatric

Note: Premixed solutions available.

IV infusion: 1.8 mg/mL, 3.6 mg/mL.

Vesicant/Extravasation Risk

Irritant with vesicant-like properties.

Storage/Stability

Tablets: Store at 20°C to 25°C (68°F to 77°F); protect from light.

Injection: Store undiluted vials and premixed solutions (Nexterone) at 20°C to 25°C (68°F to 77°F); excursions are permitted between 15°C and 30°C (59°F and 86°F). Protect from light during storage; protect from excessive heat. There is no need to protect solutions from light during administration. When vial contents are admixed in D5W to a final concentration of 1-6 mg/mL, amiodarone is stable for 24 hours in glass or polyolefin bottles and for 2 hours in polyvinyl chloride (PVC) bags; do not use evacuated glass containers as buffer may cause precipitation. Nexterone is available as premixed solutions. Although amiodarone adsorbs to PVC tubing, all clinical studies used PVC tubing and the recommended doses account for adsorption; in adults, PVC tubing is recommended. Discard any unused portions of premixed solutions.

Use: Labeled Indications

Ventricular arrhythmias: Management of life-threatening recurrent ventricular fibrillation (VF) or recurrent hemodynamically unstable ventricular tachycardia (VT) refractory to other antiarrhythmic agents or in patients intolerant of other agents used for these conditions

Use: Off-Label: Adult

Atrial fibrillation; Electrical storm and incessant ventricular tachycardia, hemodynamically stable; Nonsustained ventricular tachycardia, symptomatic; Pharmacologic conversion of atrial fibrillation to and maintenance of normal sinus rhythm; Prevention of implantable cardioverter defibrillator shocks; Prevention of postoperative atrial fibrillation and atrial flutter associated with cardiothoracic surgery; Primary prevention of sudden cardiac death due to ventricular arrhythmias; Supraventricular tachycardia (eg, atrioventricular nodal reentrant tachycardia, atrioventricular reentrant tachycardia, focal atrial tachycardia); Sustained monomorphic ventricular tachycardia, hemodynamically stable; Ventricular premature beats, symptomatic

Medication Safety Issues

Sound-alike/look-alike issues:

Amiodarone may be confused with amantadine, amiloride, inamrinone.

Coronone may be confused with Cardura, Cordran.

Nexterone may be confused with Cardene IV.

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication among its list of drug classes (antiarrhythmic agent, IV) which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Acute Care Settings ©ISMP 2024).

Older Adult: High-Risk Medication:

Beers Criteria: Amiodarone is identified in the Beers Criteria as a potentially inappropriate medication to be avoided in patients 65 years and older as first-line therapy for atrial fibrillation (unless patient has concomitant heart failure or substantial left ventricular hypertrophy) due to increased toxicities compared to other antiarrhythmics used in atrial fibrillation (Beers Criteria [AGS 2023]).

Amiodarone is identified in the Screening Tool of Older Person's Prescriptions (STOPP) criteria as a potentially inappropriate medication in older adults (≥65 years of age) with known QTc prolongation or as first-line therapy for supraventricular tachyarrhythmias (O'Mahony 2023).

International Issues:

Amyben [brand name for amiodarone (Great Britain)] may be confused with Ambien [US, Argentina, Israel]

Metabolism/Transport Effects

Substrate of CYP1A2 (Minor), CYP2C19 (Minor), CYP2C8 (Minor), CYP2D6 (Minor), CYP3A4 (Major), P-glycoprotein (Minor); **Note:** Assignment of Major/Minor substrate status based on clinically relevant drug interaction potential; **Inhibits** CYP2C9 (Weak), CYP2D6 (Weak), CYP3A4 (Weak), P-glycoprotein

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within "CYP3A4 Inducers [Strong]" are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the "Launch drug interactions program" link above.

Afatinib: P-glycoprotein/ABCB1 Inhibitors may increase serum concentrations of Afatinib. Management: If combined, administer the P-gp inhibitor simultaneously with, or after, the dose of afatinib.

Monitor closely for signs and symptoms of afatinib toxicity and if the combination is not tolerated, reduce the afatinib dose by 10 mg. *Risk D: Consider Therapy Modification*

Alfuzosin: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Aliskiren: P-glycoprotein/ABCB1 Inhibitors may increase serum concentrations of Aliskiren. *Risk C: Monitor*

ALPRAZolam: CYP3A4 Inhibitors (Weak) may increase serum concentrations of ALPRAZolam. *Risk C: Monitor*

Amifostine: Blood Pressure Lowering Agents may increase hypotensive effects of Amifostine. Management: When used at chemotherapy doses, hold blood pressure lowering medications for 24 hours before amifostine administration. If blood pressure lowering therapy cannot be held, do not administer amifostine. Use caution with radiotherapy doses of amifostine. *Risk D: Consider Therapy Modification*

Aminolevulinic Acid (Systemic): Photosensitizing Agents may increase photosensitizing effects of Aminolevulinic Acid (Systemic). *Risk X: Avoid*

Aminolevulinic Acid (Topical): Photosensitizing Agents may increase photosensitizing effects of Aminolevulinic Acid (Topical). *Risk C: Monitor*

Amisulpride (Oral): QT-prolonging Agents (Highest Risk) may increase QTc-prolonging effects of Amisulpride (Oral). Management: Consider alternatives to this drug combination. If combined, monitor for QTc interval prolongation and ventricular arrhythmias. Patients with additional risk factors for QTc prolongation may be at even greater risk. *Risk D: Consider Therapy Modification*

Antipsychotic Agents (Second Generation [Atypical]): Blood Pressure Lowering Agents may increase hypotensive effects of Antipsychotic Agents (Second Generation [Atypical]). *Risk C: Monitor*

Apixaban: Amiodarone may increase serum concentrations of Apixaban. *Risk C: Monitor*

Arginine: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Atazanavir: May increase serum concentrations of Amiodarone. *Risk X: Avoid*

Atorvastatin: Amiodarone may increase serum concentrations of Atorvastatin. *Risk C: Monitor*

Autumn Crocus: P-glycoprotein/ABCB1 Inhibitors may increase serum concentrations of Autumn Crocus. Specifically, concentrations of the active constituent colchicine may be increased *Risk X: Avoid*

Azithromycin (Systemic): QT-prolonging Agents (Highest Risk) may increase QTc-prolonging effects of Azithromycin (Systemic). Management: Consider alternatives to this combination. Patients with other risk factors (eg, older age, female sex, bradycardia, hypokalemia, hypomagnesemia, heart disease, and higher drug concentrations) are likely at greater risk for these toxicities. *Risk D: Consider Therapy Modification*

Barbiturates: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Benperidol: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Beta-Blockers: Amiodarone may increase bradycardic effects of Beta-Blockers. Possibly to the point of cardiac arrest. Amiodarone may increase serum concentrations of Beta-Blockers. *Risk C: Monitor*

Bilastine: P-glycoprotein/ABCB1 Inhibitors may increase serum concentrations of Bilastine. Management: Consider alternatives, as product labeling states that the use of bilastine with P-gp inhibitors is not recommended. Bilastine therapy should be strictly avoided in patients with moderate to severe renal insufficiency receiving P-gp inhibitors. *Risk D: Consider Therapy Modification*

Bile Acid Sequestrants: May decrease bioavailability of Amiodarone. Management: Consider alternatives to this combination due to the risk of subtherapeutic amiodarone serum concentrations. If amiodarone is coadministered with colestevlam, administer amiodarone at least 4 hours before colestevlam. *Risk D: Consider Therapy Modification*

Blood Pressure Lowering Agents: May increase hypotensive effects of Hypotension-Associated Agents. *Risk C: Monitor*

Bradycardia-Causing Agents: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Brimonidine (Topical): May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Bromperidol: May decrease hypotensive effects of Blood Pressure Lowering Agents. Blood Pressure Lowering Agents may increase hypotensive effects of Bromperidol. *Risk X: Avoid*

Calcium Channel Blockers (Nondihydropyridine): May increase bradycardic effects of Amiodarone. Calcium Channel Blockers (Nondihydropyridine) may increase serum concentrations of Amiodarone. *Risk C: Monitor*

Carbamazepine: Amiodarone may increase serum concentrations of Carbamazepine. Carbamazepine may decrease serum concentrations of Amiodarone. Carbamazepine may decrease active metabolite exposure of Amiodarone. *Risk C: Monitor*

Carbetocin: May increase QTc-prolonging effects of QT-prolonging Agents (Highest Risk). Management: Consider alternatives to this drug combination. If combined, monitor for QTc interval prolongation and ventricular arrhythmias. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

Cardiac Glycosides: Amiodarone may increase serum concentrations of Cardiac Glycosides. Management: Reduce the dose of cardiac glycosides by 30% to 50% or reduce the frequency of administration when initiating concomitant amiodarone therapy. Monitor for increased serum concentrations and toxic effects of cardiac glycosides. *Risk D: Consider Therapy Modification*

Celiprolol: P-glycoprotein/ABCB1 Inhibitors may increase serum concentrations of Celiprolol. *Risk C: Monitor*

Ceritinib: Amiodarone may increase QTc-prolonging effects of Ceritinib. Amiodarone may increase bradycardic effects of Ceritinib. Management: Avoid concomitant use if possible. Patients with other risk factors (eg, older age, female sex, bradycardia, hypokalemia, hypomagnesemia, heart disease, and higher drug concentrations) are likely at greater risk for these toxicities. *Risk D: Consider Therapy Modification*

Chloroquine: QT-prolonging Agents (Highest Risk) may increase QTc-prolonging effects of Chloroquine. Management: Consider alternatives to this combination. If combined, monitor for QTc interval prolongation and ventricular arrhythmias. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

Chlorprothixene: May increase QTc-prolonging effects of Antiarrhythmic Agents (Class III). *Risk X: Avoid*

Cimetidine: May increase serum concentrations of Amiodarone. *Risk X: Avoid*

Citalopram: QT-prolonging Agents (Highest Risk) may increase QTc-prolonging effects of Citalopram. *Risk X: Avoid*

Clarithromycin: QT-prolonging Agents (Highest Risk) may increase QTc-prolonging effects of Clarithromycin. *Risk X: Avoid*

Clofazimine: QT-prolonging Agents (Highest Risk) may increase QTc-prolonging effects of Clofazimine. Management: Consider alternatives to this combination. If combined, monitor for QTc interval prolongation and ventricular arrhythmias. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

Clomipramine: May increase QTc-prolonging effects of QT-prolonging Agents (Highest Risk). Management: Consider alternatives to this drug combination. If combined, monitor for QTc interval prolongation and ventricular arrhythmias. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

CloZAPine: Amiodarone may increase QTc-prolonging effects of CloZAPine. Amiodarone may increase serum concentrations of CloZAPine. Management: Consider alternatives to this combination. Patients with other risk factors (eg, older age, female sex, bradycardia, hypokalemia, hypomagnesemia, heart disease, and higher drug concentrations) are likely at greater risk for these toxicities. *Risk D: Consider Therapy Modification*

Colchicine Containing Products: P-glycoprotein/ABCB1 Inhibitors may increase serum concentrations of Colchicine Containing Products. Colchicine distribution into certain tissues (e.g., brain) may also be increased. Management: This combination is often contraindicated, but combined use may be permitted with dose adjustment and monitoring. Recommendations vary based on brand, indication, use of CYP3A4 inhibitors, and hepatic/renal function. See interaction monograph for details. *Risk D: Consider Therapy Modification*

CycloPHOSphamide: Adverse/toxic effects of CycloPHOSphamide and Amiodarone may increase with coadministration. Specifically, the risk of pulmonary toxicity may be enhanced. *Risk C: Monitor*

CycloSPORINE (Systemic): CYP3A4 Inhibitors (Weak) may increase serum concentrations of CycloSPORINE (Systemic). *Risk C: Monitor*

CycloSPORINE (Systemic): P-glycoprotein/ABCB1 Inhibitors may increase serum concentrations of CycloSPORINE (Systemic). *Risk C: Monitor*

CYP2C8 Inhibitors (Strong): May increase serum concentrations of Amiodarone. *Risk C: Monitor*

CYP3A4 Inducers (Strong): May decrease active metabolite exposure of Amiodarone. CYP3A4 Inducers (Strong) may decrease serum concentrations of Amiodarone. *Risk C: Monitor*

CYP3A4 Inhibitors (Moderate): May increase serum concentrations of Amiodarone. *Risk C: Monitor*

CYP3A4 Inhibitors (Strong): May increase serum concentrations of Amiodarone. Management: Consider alternatives to use of amiodarone and strong CYP3A4 inhibitors. If combined, monitor for increased amiodarone concentrations and toxicities. *Risk D: Consider Therapy Modification*

Dabigatran Etexilate: P-glycoprotein/ABCB1 Inhibitors may increase active metabolite exposure of Dabigatran Etexilate. Specifically, dabigatran concentrations may be increased. *Risk C: Monitor*

Dabrafenib: May increase QTc-prolonging effects of QT-prolonging Agents (Highest Risk). Management: Consider alternatives to this drug combination. If combined, monitor for QTc interval prolongation and ventricular arrhythmias. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

Daclatasvir: May increase bradycardic effects of Amiodarone. *Risk X: Avoid*

Dasatinib: QT-prolonging Agents (Highest Risk) may increase QTc-prolonging effects of Dasatinib. Management: Consider alternatives to this combination. Patients with other risk factors (eg, older age, female sex, bradycardia, hypokalemia, hypomagnesemia, heart disease, and higher drug concentrations) are likely at greater risk for these toxicities. *Risk D: Consider Therapy Modification*

Diazoxide (Immediate Release): May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Dihydrocodeine: Amiodarone may decrease active metabolite exposure of Dihydrocodeine. *Risk C: Monitor*

Domperidone: QT-prolonging Agents (Highest Risk) may increase QTc-prolonging effects of Domperidone. *Risk X: Avoid*

Doxepin-Containing Products: QT-prolonging Agents (Highest Risk) may increase QTc-prolonging effects of Doxepin-Containing Products. Management: Consider alternatives to this combination. Patients with other risk factors (eg, older age, female sex, bradycardia, hypokalemia, hypomagnesemia, heart disease, and higher drug concentrations) are likely at greater risk for these toxicities. *Risk D: Consider Therapy Modification*

DOXOrubicin (Conventional): P-glycoprotein/ABCB1 Inhibitors may increase serum concentrations of DOXOrubicin (Conventional). *Risk X: Avoid*

DOXOrubicin (Liposomal): P-glycoprotein/ABCB1 Inhibitors may increase serum concentrations of DOXOrubicin (Liposomal). *Risk C: Monitor*

DroPERidol: QT-prolonging Agents (Highest Risk) may increase QTc-prolonging effects of DroPERidol. Management: Consider alternatives to this combination. Patients with other risk factors (eg, older age, female sex, bradycardia, hypokalemia, hypomagnesemia, heart disease, and higher drug concentrations) are likely at greater risk for these toxicities. *Risk D: Consider Therapy Modification*

DULoxetine: Blood Pressure Lowering Agents may increase hypotensive effects of DULoxetine. *Risk C: Monitor*

Edoxaban: P-glycoprotein/ABCB1 Inhibitors may increase serum concentrations of Edoxaban. *Risk C: Monitor*

Encorafenib: May increase QTc-prolonging effects of Amiodarone. Encorafenib may decrease serum concentrations of Amiodarone. Management: Consider alternatives to this drug combination. If combined, monitor for QTc interval prolongation, ventricular arrhythmias, and decreased amiodarone concentrations. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

Ensartinib: P-glycoprotein/ABCB1 Inhibitors may increase serum concentrations of Ensartinib. *Risk X: Avoid*

Entrectinib: May increase QTc-prolonging effects of QT-prolonging Agents (Highest Risk). *Risk X: Avoid*

Erythromycin (Systemic): Amiodarone may increase QTc-prolonging effects of Erythromycin (Systemic). Erythromycin (Systemic) may increase QTc-prolonging effects of Amiodarone. Erythromycin (Systemic) may increase serum concentrations of Amiodarone. *Risk X: Avoid*

Escitalopram: QT-prolonging Agents (Highest Risk) may increase QTc-prolonging effects of Escitalopram. Management: Consider alternatives to this combination. Patients with other risk factors (eg, older age, female sex, bradycardia, hypokalemia, hypomagnesemia, heart disease, and higher drug concentrations) are likely at greater risk for these toxicities. *Risk D: Consider Therapy Modification*

Etelcalcetide: May increase QTc-prolonging effects of QT-prolonging Agents (Highest Risk). Management: Consider alternatives to this drug combination. If combined, monitor for QTc interval prolongation and ventricular arrhythmias. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

Etoposide Phosphate: P-glycoprotein/ABCB1 Inhibitors may increase serum concentrations of Etoposide Phosphate. *Risk C: Monitor*

Etoposide: P-glycoprotein/ABCB1 Inhibitors may increase serum concentrations of Etoposide. *Risk C: Monitor*

Etrasimod: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Everolimus: P-glycoprotein/ABCB1 Inhibitors may increase serum concentrations of Everolimus. *Risk C: Monitor*

Fexinidazole: May increase QTc-prolonging effects of QT-prolonging Agents (Highest Risk). *Risk X: Avoid*

Finerenone: CYP3A4 Inhibitors (Weak) may increase serum concentrations of Finerenone. *Risk C: Monitor*

Fingolimod: May increase QTc-prolonging effects of Amiodarone. *Risk X: Avoid*

Flecainide: Amiodarone may increase QTc-prolonging effects of Flecainide. Amiodarone may increase serum concentrations of Flecainide. Management: Decrease flecainide dose by 50%. Monitor for QTc interval prolongation and ventricular arrhythmias, and consider monitoring for elevated flecainide concentrations. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

Flibanserin: CYP3A4 Inhibitors (Weak) may increase serum concentrations of Flibanserin. *Risk C: Monitor*

Fluorouracil Products: QT-prolonging Agents (Highest Risk) may increase QTc-prolonging effects of Fluorouracil Products. Management: Consider alternatives to this drug combination. If combined, monitor for QTc interval prolongation and ventricular arrhythmias. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

Flupentixol: QT-prolonging Agents (Highest Risk) may increase QTc-prolonging effects of Flupentixol. *Risk X: Avoid*

Fosphenytoin-Phenytoin: May decrease serum concentrations of Amiodarone. Amiodarone may increase serum concentrations of Fosphenytoin-Phenytoin. *Risk C: Monitor*

Fusidic Acid (Systemic): May increase serum concentrations of CYP3A4 Substrates (High risk with Inhibitors). Management: Consider avoiding this combination if possible. If required, monitor patients closely for increased adverse effects of the CYP3A4 substrate. *Risk D: Consider Therapy Modification*

Gadobenate Dimeglumine: QT-prolonging Agents (Highest Risk) may increase QTc-prolonging effects of Gadobenate Dimeglumine. Management: Consider alternatives to this combination. If combined, monitor for QTc interval prolongation and ventricular arrhythmias. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

Gemifloxacin: QTc-prolonging effects of Gemifloxacin and Amiodarone may increase with coadministration. *Risk X: Avoid*

Gilteritinib: May increase QTc-prolonging effects of QT-prolonging Agents (Highest Risk). Management: Consider alternatives to this combination. If use is necessary, monitor for QTc interval prolongation and arrhythmias. *Risk D: Consider Therapy Modification*

Givinostat: May increase QTc-prolonging effects of QT-prolonging Agents (Highest Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Glecaprevir and Pibrentasvir: P-glycoprotein/ABCB1 Inhibitors may increase serum concentrations of Glecaprevir and Pibrentasvir. *Risk C: Monitor*

Grapefruit Juice: May increase serum concentrations of Amiodarone. *Risk X: Avoid*

Halofantrine: QT-prolonging Agents (Highest Risk) may increase QTc-prolonging effects of Halofantrine. Management: Consider alternatives to this combination. If combined, monitor for QTc interval prolongation and ventricular arrhythmias. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

Haloperidol: Amiodarone may increase QTc-prolonging effects of Haloperidol. Management: Consider alternatives to this drug combination. If combined, monitor for QTc interval prolongation and ventricular arrhythmias. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

Herbal Products with Blood Pressure Lowering Effects: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Hydroxyzine: May increase QTc-prolonging effects of QT-prolonging Agents (Highest Risk). *Risk C: Monitor*

Hypotension-Associated Agents: Blood Pressure Lowering Agents may increase hypotensive effects of Hypotension-Associated Agents. *Risk C: Monitor*

Iloperidone: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Imipramine: May increase QTc-prolonging effects of QT-prolonging Agents (Highest Risk). Management: Consider alternatives to this drug combination. If combined, monitor for QTc interval prolongation and ventricular arrhythmias. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

Inotuzumab Ozogamicin: QT-prolonging Agents (Highest Risk) may increase QTc-prolonging effects of Inotuzumab Ozogamicin. Management: Consider alternatives to this combination. If combined, monitor for QTc interval prolongation and ventricular arrhythmias. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

Ivabradine: Bradycardia-Causing Agents may increase bradycardic effects of Ivabradine. *Risk C: Monitor*

Ixabepilone: CYP3A4 Inhibitors (Weak) may increase serum concentrations of Ixabepilone. *Risk C: Monitor*

Lacosamide: Antiarrhythmic Agents (Class III) may increase adverse/toxic effects of Lacosamide. Specifically the risk for bradycardia, ventricular tachyarrhythmias, or a prolonged PR interval may be increased. *Risk C: Monitor*

Lacosamide: Bradycardia-Causing Agents may increase AV-blocking effects of Lacosamide. *Risk C: Monitor*

Landiolol: Bradycardia-Causing Agents may increase bradycardic effects of Landiolol. *Risk X: Avoid*

Lapatinib: P-glycoprotein/ABCB1 Inhibitors may increase serum concentrations of Lapatinib. *Risk C: Monitor*

Larotrectinib: P-glycoprotein/ABCB1 Inhibitors may increase serum concentrations of Larotrectinib. *Risk C: Monitor*

Lefamulin: May increase QTc-prolonging effects of QT-prolonging CYP3A4 Substrates. Management: Do not use lefamulin tablets with QT-prolonging CYP3A4 substrates. Lefamulin prescribing information lists this combination as contraindicated. *Risk X: Avoid*

Lemborexant: CYP3A4 Inhibitors (Weak) may increase serum concentrations of Lemborexant. Management: The maximum recommended dosage of lemborexant is 5 mg, no more than once per night, when coadministered with weak CYP3A4 inhibitors. *Risk D: Consider Therapy Modification*

Levodopa-Foslevodopa: Blood Pressure Lowering Agents may increase hypotensive effects of Levodopa-Foslevodopa. *Risk C: Monitor*

Levofloxacin-Containing Products (Systemic): May increase QTc-prolonging effects of Amiodarone. *Risk X: Avoid*

Levoketoconazole: QT-prolonging CYP3A4 Substrates may increase QTc-prolonging effects of Levoketoconazole. Levoketoconazole may increase serum concentrations of QT-prolonging CYP3A4 Substrates. *Risk X: Avoid*

Lidocaine (Systemic): Amiodarone may increase serum concentrations of Lidocaine (Systemic). *Risk C: Monitor*

Lofepamine: May increase arrhythmogenic effects of Amiodarone. *Risk X: Avoid*

Lofexidine: May increase QTc-prolonging effects of Amiodarone. Management: Consider alternatives to this combination. Patients with other risk factors (eg, older age, female sex, bradycardia, hypokalemia, hypomagnesemia, heart disease, and higher drug concentrations) are likely at greater risk for these toxicities. *Risk D: Consider Therapy Modification*

Lomitapide: CYP3A4 Inhibitors (Weak) may increase serum concentrations of Lomitapide. Management: Decrease lomitapide dose 50% if starting a weak CYP3A4 inhibitor. Lomitapide dose may be up-titrated, but the max recommended dose is lower - 10 mg for patients aged 2 to 10 yrs; 20 mg for patients aged 11 to 17 yrs; 30 mg for patients 18 yrs and older. *Risk D: Consider Therapy Modification*

Lonafarnib: May increase QTc-prolonging effects of QT-prolonging Agents (Highest Risk). Management: Consider alternatives to this drug combination. If combined, monitor for QTc interval prolongation and ventricular arrhythmias. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

Loratadine: Amiodarone may increase serum concentrations of Loratadine. Management: Due to reported QT interval prolongation and Torsades de Pointes with this combination, consider an alternative to loratadine when possible. If concomitant use cannot be avoided, monitor QT interval and for signs of dysrhythmias (eg, syncope). *Risk D: Consider Therapy Modification*

Lormetazepam: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Lovastatin: Amiodarone may increase serum concentrations of Lovastatin. Management: Consider using a non-interacting statin (pravastatin, pitavastatin) in patients on amiodarone. If combined, limit the lovastatin dose to 40 mg daily and monitor for lovastatin toxicities (eg, myalgia, liver function test elevations, rhabdomyolysis). *Risk D: Consider Therapy Modification*

Macitentan: Amiodarone may increase serum concentrations of Macitentan. *Risk C: Monitor*

Mavoxixafor: P-glycoprotein/ABCB1 Inhibitors may increase serum concentrations of Mavoxixafor. *Risk C: Monitor*

Meglumine Antimoniate: May increase QTc-prolonging effects of QT-prolonging Agents (Highest Risk). Management: Consider alternatives to this drug combination. If combined, monitor for QTc interval prolongation and ventricular arrhythmias. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

Meloxicam: Amiodarone may increase serum concentrations of Meloxicam. *Risk C: Monitor*

Metergoline: May decrease antihypertensive effects of Blood Pressure Lowering Agents. Blood Pressure Lowering Agents may increase orthostatic hypotensive effects of Metergoline. *Risk C: Monitor*

Methadone: Amiodarone may increase QTc-prolonging effects of Methadone. Management: Consider alternatives to this drug combination. If combined, monitor for QTc interval prolongation and ventricular arrhythmias. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

Methoxsalen (Systemic): Photosensitizing Agents may increase photosensitizing effects of Methoxsalen (Systemic). *Risk C: Monitor*

Methylergonovine: CYP3A4 Inhibitors (Weak) may increase serum concentrations of Methylergonovine. *Risk C: Monitor*

Midazolam: CYP3A4 Inhibitors (Weak) may increase serum concentrations of Midazolam. *Risk C: Monitor*

Midodrine: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Midostaurin: QT-prolonging Agents (Highest Risk) may increase QTc-prolonging effects of Midostaurin. Management: Consider alternatives to this combination. If combined, monitor for QTc interval prolongation and ventricular arrhythmias. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

Milsaperidone: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Molsidomine: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Morphine (Systemic): P-glycoprotein/ABCB1 Inhibitors may increase serum concentrations of Morphine (Systemic). *Risk C: Monitor*

Moxifloxacin (Systemic): QT-prolonging Agents (Highest Risk) may increase QTc-prolonging effects of Moxifloxacin (Systemic). *Risk X: Avoid*

Nadolol: P-glycoprotein/ABCB1 Inhibitors may increase serum concentrations of Nadolol. *Risk C: Monitor*

Naftopidil: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Naldemedine: P-glycoprotein/ABCB1 Inhibitors may increase serum concentrations of Naldemedine. *Risk C: Monitor*

Naloxegol: P-glycoprotein/ABCB1 Inhibitors may increase serum concentrations of Naloxegol. *Risk C: Monitor*

Nelfinavir: May increase serum concentrations of Amiodarone. *Risk X: Avoid*

Nicergoline: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nicorandil: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nilotinib: QT-prolonging Agents (Highest Risk) may increase QTc-prolonging effects of Nilotinib. *Risk X: Avoid*

NiMODipine: CYP3A4 Inhibitors (Weak) may increase serum concentrations of NiMODipine. *Risk C: Monitor*

Nirmatrelvir and Ritonavir: May increase serum concentrations of Amiodarone. *Risk X: Avoid*

Nitroprusside: Blood Pressure Lowering Agents may increase hypotensive effects of Nitroprusside. *Risk C: Monitor*

Obinutuzumab: May increase hypotensive effects of Blood Pressure Lowering Agents. Management: Consider temporarily withholding blood pressure lowering medications beginning 12 hours prior to obinutuzumab infusion and continuing until 1 hour after the end of the infusion. *Risk D: Consider Therapy Modification*

OLANzapine: QT-prolonging Agents (Highest Risk) may increase QTc-prolonging effects of OLANzapine. Management: Consider alternatives to this combination. Patients with other risk factors (eg, older age, female sex, bradycardia, hypokalemia, hypomagnesemia, heart disease, and higher drug concentrations) are likely at greater risk for these toxicities. *Risk D: Consider Therapy Modification*

Ondansetron: Amiodarone may increase QTc-prolonging effects of Ondansetron. Management: Consider alternatives to this combination. If combined, monitor for QTc interval prolongation and ventricular arrhythmias. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

Orlistat: May decrease serum concentrations of Amiodarone. *Risk C: Monitor*

Osimertinib: QT-prolonging Agents (Highest Risk) may increase QTc-prolonging effects of Osimertinib. Management: Consider alternatives to this combination. Patients with other risk factors (eg, older age, female sex, bradycardia, hypokalemia, hypomagnesemia, heart disease, and higher drug concentrations) are likely at greater risk for these toxicities. *Risk D: Consider Therapy Modification*

Oxygen: May increase adverse/toxic effects of Amiodarone. Specifically, amiodarone pulmonary toxicity may be enhanced. *Risk C: Monitor*

Oxytocin: May increase QTc-prolonging effects of QT-prolonging Agents (Highest Risk). Management: Consider alternatives to this drug combination. If combined, monitor for QTc interval prolongation and ventricular arrhythmias. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

Ozanimod: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Pacritinib: QT-prolonging Agents (Highest Risk) may increase QTc-prolonging effects of Pacritinib. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

PAZOPanib: Amiodarone may increase QTc-prolonging effects of PAZOPanib. Amiodarone may increase serum concentrations of PAZOPanib. *Risk X: Avoid*

Pentamidine (Systemic): QT-prolonging Agents (Highest Risk) may increase QTc-prolonging effects of Pentamidine (Systemic). Management: Consider alternatives to this combination. If combined, monitor for QTc interval prolongation and ventricular arrhythmias. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

Pentoxifylline: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Pholcodine: Blood Pressure Lowering Agents may increase hypotensive effects of Pholcodine. *Risk C: Monitor*

Phosphodiesterase 5 Inhibitors: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Pilsicainide: QT-prolonging Agents (Highest Risk) may increase QTc-prolonging effects of Pilsicainide. Management: Consider alternatives to this combination. If combined, monitor for QTc interval prolongation and ventricular arrhythmias. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

Pimozide: QT-prolonging Agents (Highest Risk) may increase QTc-prolonging effects of Pimozide. Management: Consider alternatives to this combination. Patients with other risk factors (eg, older age, female sex, bradycardia, hypokalemia, hypomagnesemia, heart disease, and higher drug concentrations) are likely at greater risk for these toxicities. *Risk X: Avoid*

Piperaquine: QT-prolonging Agents (Highest Risk) may increase QTc-prolonging effects of Piperaquine. *Risk X: Avoid*

Ponesimod: Bradycardia-Causing Agents may increase bradycardic effects of Ponesimod. Management: Avoid coadministration of ponesimod with drugs that may cause bradycardia when possible. If combined, monitor heart rate closely and consider obtaining a cardiology consult. Do not initiate ponesimod in patients on beta-blockers if HR is less than 55 bpm. *Risk D: Consider Therapy Modification*

Porfimer: Photosensitizing Agents may increase photosensitizing effects of Porfimer. *Risk X: Avoid*

Posaconazole: May increase serum concentrations of QT-prolonging CYP3A4 Substrates. Posaconazole may increase QTc-prolonging effects of QT-prolonging CYP3A4 Substrates. *Risk X: Avoid*

Pralsetinib: P-glycoprotein/ABCB1 Inhibitors may increase serum concentrations of Pralsetinib. Management: If this combo cannot be avoided, decrease pralsetinib dose from 400 mg daily to 300 mg daily; from 300 mg daily to 200 mg daily; and from 200 mg daily to 100 mg daily. *Risk D: Consider Therapy Modification*

Probucol: QT-prolonging Agents (Highest Risk) may increase QTc-prolonging effects of Probucol. *Risk X: Avoid*

Propafenone: Amiodarone may increase QTc-prolonging effects of Propafenone. *Risk X: Avoid*

Propofol: May increase QTc-prolonging effects of QT-prolonging Agents (Highest Risk). Management: Consider alternatives to this drug combination. If combined, monitor for QTc interval prolongation and ventricular arrhythmias. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

Prostacyclin Analogues: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

QT-prolonging Agents (Indeterminate Risk - Avoid): May increase QTc-prolonging effects of QT-prolonging Agents (Highest Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Agents (Indeterminate Risk - Caution): May increase QTc-prolonging effects of QT-prolonging Agents (Highest Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Class IA Antiarrhythmics (Highest Risk): May increase QTc-prolonging effects of Amiodarone. Management: Consider alternatives to this drug combination. If combined, consider dose reductions of the class IA antiarrhythmic (30% to 50%) and monitor for QTc interval prolongation and ventricular arrhythmias. *Risk D: Consider Therapy Modification*

QT-prolonging Class III Antiarrhythmics (Highest Risk): May increase QTc-prolonging effects of Amiodarone. Management: Consider alternatives to this drug combination. If combined, monitor for QTc interval prolongation and ventricular arrhythmias. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

QT-Prolonging Inhalational Anesthetics (Moderate Risk): May increase QTc-prolonging effects of QT-prolonging Agents (Highest Risk). Management: Consider alternatives to this drug combination. If combined, monitor for QTc interval prolongation and ventricular arrhythmias. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

QT-prolonging Kinase Inhibitors (Highest Risk): May increase QTc-prolonging effects of Amiodarone. Management: Consider alternatives to this drug combination. If combined, monitor for QTc interval prolongation and ventricular arrhythmias. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

QT-prolonging Miscellaneous Agents (Highest Risk): May increase QTc-prolonging effects of Amiodarone. Management: Consider alternatives to this drug combination. If combined, monitor for QTc interval prolongation and ventricular arrhythmias. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

QT-prolonging Moderate CYP3A4 Inhibitors (Moderate Risk): May increase QTc-prolonging effects of Amiodarone. QT-prolonging Moderate CYP3A4 Inhibitors (Moderate Risk) may increase serum concentrations of Amiodarone. Management: Consider alternatives to this drug combination. If combined, monitor for QTc interval prolongation and ventricular arrhythmias. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

QT-prolonging Strong CYP3A4 Inhibitors (Highest Risk): May increase QTc-prolonging effects of Amiodarone. QT-prolonging Strong CYP3A4 Inhibitors (Highest Risk) may increase serum concentrations of Amiodarone. Management: Consider alternatives to this combination. If combined, monitor for increased amiodarone toxicities, including QTc interval prolongation and ventricular arrhythmias. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

QUetiapine: QT-prolonging Agents (Highest Risk) may increase QTc-prolonging effects of QUetiapine. *Risk X: Avoid*

Quizartinib: May increase QTc-prolonging effects of QT-prolonging Agents (Highest Risk). Management: Consider alternatives to this combination. Patients with other risk factors (eg, older age, female sex, bradycardia, hypokalemia, hypomagnesemia, heart disease, and higher drug concentrations) are likely at greater risk for these toxicities. *Risk D: Consider Therapy Modification*

Ranolazine: P-glycoprotein/ABCB1 Inhibitors may increase serum concentrations of Ranolazine. *Risk C: Monitor*

Rasagiline: Blood Pressure Lowering Agents may increase hypotensive effects of Rasagiline. *Risk C: Monitor*

Red Yeast Rice: Amiodarone may increase serum concentrations of Red Yeast Rice. Management: Consider using a non-interacting statin (eg, pravastatin, pitavastatin) in patients on amiodarone. If combined, limit the adult red yeast rice dose to the equivalent of lovastatin 40 mg daily and monitor for toxicities (eg, myalgia, rhabdomyolysis). *Risk D: Consider Therapy Modification*

Relugolix, Estradiol, and Norethindrone: P-glycoprotein/ABCB1 Inhibitors may increase serum concentrations of Relugolix, Estradiol, and Norethindrone. Management: Avoid use of relugolix/estradiol/norethindrone with P-glycoprotein (P-gp) inhibitors. If concomitant use is unavoidable, relugolix/estradiol/norethindrone should be administered at least 6 hours before the P-gp inhibitor. *Risk D: Consider Therapy Modification*

Relugolix: P-glycoprotein/ABCB1 Inhibitors may increase serum concentrations of Relugolix. Management: Avoid coadministration of relugolix with oral P-gp inhibitors whenever possible. If combined, take relugolix at least 6 hours prior to the P-gp inhibitor and monitor patients more frequently for adverse reactions. *Risk D: Consider Therapy Modification*

Remdesivir: Hepatotoxic effects of Remdesivir and Amiodarone may increase with coadministration. Bradycardic effects of Remdesivir and Amiodarone may increase with coadministration. Remdesivir may increase QTc-prolonging effects of Amiodarone. Management: Consider alternatives to this combination. Patients with other risk factors (eg, older age, female sex, bradycardia, hypokalemia, hypomagnesemia, heart disease, and higher drug concentrations) are likely at greater risk for these toxicities. *Risk D: Consider Therapy Modification*

Repotrectinib: P-glycoprotein/ABCB1 Inhibitors may increase serum concentrations of Repotrectinib. *Risk X: Avoid*

Ribociclib: QT-prolonging Agents (Highest Risk) may increase QTc-prolonging effects of Ribociclib. *Risk X: Avoid*

RifAXIMin: P-glycoprotein/ABCB1 Inhibitors may increase serum concentrations of RifAXIMin. *Risk C: Monitor*

Rimegepant: P-glycoprotein/ABCB1 Inhibitors may increase serum concentrations of Rimegepant. Management: Avoid administration of another dose of rimegepant within 48 hours if given concomitantly with a P-glycoprotein (P-gp) inhibitor. *Risk D: Consider Therapy Modification*

RisperiDONE: QT-prolonging Agents (Highest Risk) may increase QTc-prolonging effects of RisperiDONE. QT-prolonging Agents (Highest Risk) may increase CNS depressant effects of RisperiDONE. Management: Consider alternatives to this drug combination. If combined, monitor for QTc interval prolongation and ventricular arrhythmias. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

Ritonavir: May increase serum concentrations of Amiodarone. Management: Ritonavir US prescribing information lists this combination as contraindicated. Amiodarone use should be avoided with lopinavir/ritonavir, but if the combination must be used, monitor closely for increased amiodarone serum concentrations and effects. *Risk X: Avoid*

Rivaroxaban: Amiodarone may increase serum concentrations of Rivaroxaban. *Risk C: Monitor*

RomiDEPsin: P-glycoprotein/ABCB1 Inhibitors may increase serum concentrations of RomiDEPsin. *Risk C: Monitor*

ROPivacaine: Antiarrhythmic Agents (Class III) may increase arrhythmogenic effects of ROPivacaine. *Risk C: Monitor*

Sebetralstat: CYP3A4 Inhibitors (Weak) may increase serum concentrations of Sebetralstat. *Risk C: Monitor*

Selegiline: Blood Pressure Lowering Agents may increase hypotensive effects of Selegiline. *Risk C: Monitor*

Sertindole: May increase QTc-prolonging effects of QT-prolonging Agents (Highest Risk). *Risk X: Avoid*

Silodosin: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Silodosin: P-glycoprotein/ABCB1 Inhibitors may increase serum concentrations of Silodosin. *Risk C: Monitor*

Simvastatin: Amiodarone may increase serum concentrations of Simvastatin. Amiodarone may increase active metabolite exposure of Simvastatin. Management: Consider using a non-interacting statin (pravastatin) in patients on amiodarone. If combined, limit the adult simvastatin dose to 20 mg daily and monitor for evidence of simvastatin toxicities (eg, myalgia, liver function test elevations, rhabdomyolysis). *Risk D: Consider Therapy Modification*

Siponimod: Bradycardia-Causing Agents may increase bradycardic effects of Siponimod. Management: Avoid coadministration of siponimod with drugs that may cause bradycardia. If combined, consider obtaining a cardiology consult regarding patient monitoring. *Risk D: Consider Therapy Modification*

Sirolimus (Conventional): P-glycoprotein/ABCB1 Inhibitors may increase serum concentrations of Sirolimus (Conventional). Management: Avoid concurrent use of sirolimus with P-glycoprotein (P-

gp) inhibitors when possible and alternative agents with lesser interaction potential with sirolimus should be considered. Monitor for increased sirolimus concentrations/toxicity if combined. *Risk D: Consider Therapy Modification*

Sirolimus (Protein Bound): P-glycoprotein/ABCB1 Inhibitors may increase serum concentrations of Sirolimus (Protein Bound). *Risk X: Avoid*

Sodium Iodide I131: Amiodarone may decrease therapeutic effects of Sodium Iodide I131. Management: Discontinue amiodarone at least 6 months before sodium iodide I-131 administration, and avoid concurrent use. *Risk X: Avoid*

Sodium Perchlorate: Amiodarone may decrease therapeutic effects of Sodium Perchlorate. *Risk C: Monitor*

Sofosbuvir: May increase bradycardic effects of Amiodarone. Management: Use alternative to a sofosbuvir-containing combo or to amiodarone when possible. If alternatives not possible, monitor in inpatient setting for first 48 hours of coadministration with daily outpatient monitoring for at least 2 weeks. *Risk D: Consider Therapy Modification*

Sparfloxacin: QT-prolonging Agents (Highest Risk) may increase QTc-prolonging effects of Sparfloxacin. *Risk X: Avoid*

Sulfonylureas: Amiodarone may increase hypoglycemic effects of Sulfonylureas. *Risk C: Monitor*

SUNITinib: QT-prolonging Agents (Highest Risk) may increase QTc-prolonging effects of SUNITinib. Management: Consider alternatives to this combination. Patients with other risk factors (eg, older age, female sex, bradycardia, hypokalemia, hypomagnesemia, heart disease, and higher drug concentrations) are likely at greater risk for these toxicities. *Risk D: Consider Therapy Modification*

Tacrolimus (Systemic): May increase QTc-prolonging effects of Amiodarone. Amiodarone may increase serum concentrations of Tacrolimus (Systemic). *Risk C: Monitor*

Talazoparib: Amiodarone may increase serum concentrations of Talazoparib. Management: In breast cancer, if concurrent use cannot be avoided, reduce talazoparib dose to 0.75 mg once daily. In prostate cancer, monitor patients for increased adverse events. *Risk D: Consider Therapy Modification*

Taletrectinib: May increase QTc-prolonging effects of QT-prolonging Agents (Highest Risk). *Risk X: Avoid*

Temoporfin: Photosensitizing Agents may increase photosensitizing effects of Temoporfin. *Risk X: Avoid*

Tenofovir Disoproxil: P-glycoprotein/ABCB1 Inhibitors may increase serum concentrations of Tenofovir Disoproxil. *Risk C: Monitor*

Terbutaline: May increase QTc-prolonging effects of QT-prolonging Agents (Highest Risk). Management: Consider alternatives to this combination. Patients with other risk factors (eg, older age, female sex, bradycardia, hypokalemia, hypomagnesemia, heart disease, and higher drug concentrations) are likely at greater risk for these toxicities. *Risk D: Consider Therapy Modification*

Thioridazine: QT-prolonging Agents (Highest Risk) may increase QTc-prolonging effects of Thioridazine. *Risk X: Avoid*

Thyroid Products: Amiodarone may decrease therapeutic effects of Thyroid Products. *Risk C: Monitor*

Tipranavir: May increase serum concentrations of Amiodarone. *Risk X: Avoid*

TiZANidine: Amiodarone may increase serum concentrations of TiZANidine. *Risk C: Monitor*

TOLBUTamide: CYP2C9 Inhibitors (Weak) may increase serum concentrations of TOLBUTamide. *Risk C: Monitor*

Tolcapone: Blood Pressure Lowering Agents may increase hypotensive effects of Tolcapone. *Risk C: Monitor*

Topotecan: P-glycoprotein/ABCB1 Inhibitors may increase serum concentrations of Topotecan. *Risk X: Avoid*

Toremifene: QT-prolonging Agents (Highest Risk) may increase QTc-prolonging effects of Toremifene. Management: Consider alternatives to this combination. Patients with other risk factors (eg, older age, female sex, bradycardia, hypokalemia, hypomagnesemia, heart disease, and higher drug concentrations) are likely at greater risk for these toxicities. *Risk D: Consider Therapy Modification*

Torsemide: Amiodarone may increase hypotensive effects of Torsemide. Amiodarone may increase serum concentrations of Torsemide. *Risk C: Monitor*

Triazolam: CYP3A4 Inhibitors (Weak) may increase serum concentrations of Triazolam. *Risk C: Monitor*

Ubrogepant: CYP3A4 Inhibitors (Weak) may increase serum concentrations of Ubrogepant. Management: In patients taking weak CYP3A4 inhibitors, the initial and second dose (given at least 2 hours later if needed) of ubrogepant should be limited to 50 mg. *Risk D: Consider Therapy Modification*

Vemurafenib: QT-prolonging Agents (Highest Risk) may increase QTc-prolonging effects of Vemurafenib. Management: Consider alternatives to this combination. Patients with other risk factors (eg, older age, female sex, bradycardia, hypokalemia, hypomagnesemia, heart disease, and higher drug concentrations) are likely at greater risk for these toxicities. *Risk D: Consider Therapy Modification*

Venetoclax: P-glycoprotein/ABCB1 Inhibitors may increase serum concentrations of Venetoclax. Management: Reduce the venetoclax dose by at least 50% in patients requiring concomitant treatment with P-glycoprotein (P-gp) inhibitors. Resume the previous venetoclax dose 2 to 3 days after discontinuation of a P-gp inhibitor. *Risk D: Consider Therapy Modification*

Verteporfin: Photosensitizing Agents may increase photosensitizing effects of Verteporfin. *Risk C: Monitor*

Vincristine: P-glycoprotein/ABCB1 Inhibitors may increase serum concentrations of Vincristine. *Risk X: Avoid*

Vitamin K Antagonists: Amiodarone may increase serum concentrations of Vitamin K Antagonists. Management: Monitor patients extra closely for evidence of increased anticoagulant effects (elevated INR) if amiodarone is started. Consider an empiric reduction of 30% to 50% in vitamin K antagonist dose. *Risk D: Consider Therapy Modification*

Voriconazole: Amiodarone may increase QTc-prolonging effects of Voriconazole. Voriconazole may increase serum concentrations of Amiodarone. *Risk X: Avoid*

Food Interactions

Food increases the rate and extent of absorption of amiodarone. Grapefruit juice increases bioavailability of oral amiodarone by 50% and decreases the conversion of amiodarone to N-DEA (active metabolite); altered effects are possible. Management: Take consistently with regard to meals; grapefruit juice should be avoided during therapy.

Pregnancy Considerations

Amiodarone and the active metabolite, N-desethylamiodarone, cross the placenta (Plomp 1992).

In utero exposure may cause fetal harm. Reported risks include neonatal bradycardia, QT prolongation, and periodic ventricular extrasystoles; neonatal hypothyroidism (with or without goiter); neonatal hyperthyroxinemia; neurodevelopmental abnormalities independent of thyroid function; jerk nystagmus with synchronous head titubation; fetal growth retardation; and/or premature birth.

Oral or IV amiodarone should be used in pregnancy only to treat arrhythmias refractory to other treatments or when other treatments are contraindicated (ACC/AHA/HRS [Page 2015]).

Amiodarone (administered either maternally or directly to the fetus) may be considered for the in utero management of fetal atrial flutter and in life-threatening cases of sustained fetal supraventricular tachycardia refractory to first and second line agents, but because of potential toxicity, risks and benefits should be assessed (AHA [Donofrio 2014]; Kang 2015).

If in utero exposure occurs, newborns should be monitored for thyroid disorders and cardiac arrhythmias.

Breastfeeding Considerations

Amiodarone and its active metabolite are present in breast milk.

Actual concentrations of amiodarone and the active metabolite in breast milk vary greatly. The relative infant dose (RID) of amiodarone is reported by the manufacturer to be between 3.5% and 45% of the weight-adjusted maternal dose.

In general, breastfeeding is considered acceptable when the RID of a medication is <10%; when the RID is >25% breastfeeding should generally be avoided (Anderson 2016; Ito 2000).

Hypothyroidism and bradycardia have been observed in breastfed infants. The manufacturer does not recommend breastfeeding during therapy. If the mother wishes to breastfeed, thyroid function and plasma concentrations of amiodarone in the infant should be monitored (Ito 2000). Due to the long half-life, amiodarone and the metabolite may be present in breast milk for weeks following discontinuation of maternal therapy (Hall 2003; Javot 2018; Khurana 2014).

Dietary Considerations

Take consistently with regard to meals. Amiodarone is a potential source of large amounts of inorganic iodine; ~3 mg of inorganic iodine per 100 mg of amiodarone is released into the systemic circulation. Recommended daily allowance for iodine in adults is 150 mcg.

Grapefruit juice is not recommended.

Monitoring Parameters

BP, heart rate (ECG) and rhythm throughout therapy; history and physical exam every 3 to 6 months; assess patient for signs of lethargy, edema of the hands or feet, weight loss, and pulmonary toxicity (baseline pulmonary function tests and chest X-ray; continue monitoring chest X-ray every 3 to 6 months during therapy); LFTs (baseline and semiannually), signs or symptoms of clinical liver injury; monitor serum electrolytes, especially potassium and magnesium. Assess thyroid function tests before initiation of treatment and then periodically thereafter (some experts suggest every 3 to 6 months; particularly in elderly patients and in patients with underlying thyroid dysfunction). If signs or symptoms of thyroid disease or arrhythmia breakthrough/exacerbation occur then immediate re-evaluation is necessary. Amiodarone partially inhibits the peripheral conversion of T_4 to T_3 ; serum T_4 and reverse T_3 concentrations may be increased and serum T_3 may be decreased; most patients remain clinically euthyroid; however, clinical hypothyroidism or hyperthyroidism may occur (Epstein 2016).

Perform regular ophthalmic exams (Epstein 2016).

Patients with implantable cardiac devices: Monitor pacing or defibrillation thresholds with initiation of amiodarone and during treatment (Epstein 2016).

Consult individual institutional policies and procedures.

Reference Range

Adults: Therapeutic: 0.5 to 2.5 mg/L (SI: 1 to 4 micromole/L) (parent); desethyl metabolite is active and is present in equal concentration to parent drug.

Mechanism of Action

Class III antiarrhythmic agent which inhibits adrenergic stimulation (alpha- and beta-blocking properties), affects sodium, potassium, and calcium channels, prolongs the action potential and refractory period in myocardial tissue; decreases AV conduction and sinus node function

Pharmacokinetics (Adult Data Unless Noted)

Onset of action: Oral: 2 days to 3 weeks; IV: (electrophysiologic effects) within hours; Antiarrhythmic effects: 2 to 3 days to 1 to 3 weeks; mean onset of effect may be shorter in children vs adults and in patients receiving IV loading doses (Coumel 1983).

Peak effect: 1 week to 5 months.

Duration after discontinuing therapy: Variable, 2 weeks to months: Children: Less than a few weeks; Adults: Several months (Coumel 1983).

Note: Duration after discontinuation may be shorter in children than adults.

Absorption: Oral: Slow and variable.

Distribution:

IV: Rapid redistribution with a decrease to 10% of peak values within 30 to 45 minutes after completion of infusion.

IV single dose: V_{dss} : Mean range: 40 to 84 L/kg.

Oral: V_d : 66 L/kg (range: 18 to 148 L/kg).

Protein binding: >96%.

Metabolism: Hepatic via CYP2C8 and 3A4 to active N-desethylamiodarone metabolite; possible enterohepatic recirculation.

Bioavailability: Oral: ~50% (range: 35% to 65%).

Half-life elimination: **Note:** Half-life is shortened in children vs adults (Coumel 1983).

Amiodarone:

Single dose: 58 days (range: 15 to 142 days).

Oral chronic therapy: Mean range: 40 to 55 days (range: 26 to 107 days).

IV single dose: Mean range: 9 to 36 days.

N-desethylamiodarone (active metabolite):

Single dose: 36 days (range: 14 to 75 days).

Oral chronic therapy: 61 days.

IV single dose: Mean range: 9 to 30 days.

Time to peak, serum: Oral: 3 to 7 hours.

Excretion: Feces; urine (<1% as unchanged drug).

Pharmacokinetics: Additional Considerations (Adult Data Unless Noted)

Hepatic function impairment: After a single dose of amiodarone injection in cirrhotic patients, C_{max} was significantly lower and average concentration values were seen for N-desethylamiodarone, but mean amiodarone levels were unchanged.

Older adult: Clearance is lower and half-life is increased.

Medication Guide and/or Vaccine Information Statement (VIS)

An FDA-approved patient medication guide, which is available with the product information and as follows, must be dispensed with this medication:

Coradarone: https://www.accessdata.fda.gov/drugsatfda_docs/label/2018/018972s054lbl.pdf#page=13

Pacerone: <https://dailymed.nlm.nih.gov/dailymed/medguide.cfm?setid=4c4ec9b0-13a5-49a8-b57b-b3f64e4316a7>

Patient Counseling Points

(For additional information see "Amiodarone: Patient drug information")

What is this drug used for?

- It is used to treat certain types of abnormal heartbeats.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

- Constipation
- Upset stomach or throwing up
- Feeling tired or weak
- Not hungry
- Stomach pain
- Headache
- Trouble sleeping
- Flushing
- Change in taste
- Change in sense of smell
- More saliva

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

All products:

- Diarrhea or throwing up
- A very bad eye problem like a change in eyesight and sometimes loss of eyesight
- Stevens-Johnson syndrome/toxic epidermal necrolysis like red, swollen, blistered, or peeling skin (with or without fever); red or irritated eyes; or sores in your mouth, throat, nose, or eyes
- Liver problems like dark urine, feeling tired, not hungry, upset stomach or stomach pain, light-colored stools, throwing up, or yellow skin or eyes
- Lung or breathing problems like shortness of breath or other trouble breathing, cough, or fever
- Thyroid problems like change in weight; feeling nervous, excitable, restless, or weak; hair thinning; depression; neck swelling; not able to focus; trouble with heat or cold; menstrual changes; shakiness; or sweating
- Change in balance
- Shakiness, trouble moving around, or stiffness
- Blue or gray skin color
- Sunburn
- Coughing up blood
- Chest pain or pressure
- A burning, numbness, or tingling feeling that is not normal
- Fast or slow heartbeat
- An abnormal heartbeat that is new or worse
- Dizziness or passing out
- Change in eyesight, eye pain, or very bad eye irritation
- If bright lights bother your eyes

- Trouble controlling body movements
- Muscle weakness
- Shortness of breath, a big weight gain, or swelling in the arms or legs
- Any unexplained bruising or bleeding
- Lowered interest in sex
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Injection:

- Redness, burning, pain, swelling, blisters, skin sores, or leaking of fluid where the drug is going into your body

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Amirone | Corðarone;
- (AR) Argentina: Amiocar | Amiodarona | Amiodarona duncan | Amiodarona fabra | Amiodarona gen med | Amiodarona lacefa | Amiodarona larjan | Amiodarona Northia | Amiodarona puntanos | Amiodarona vannier | Angoten | Angoten anguten | Atlansil | Batmotropin | Cistimela | Coronovo | Miodarona | Miotenk | Ritmocardyl | Ritmonil;
- (AT) Austria: Amiodaron | Amiodaron hameln | Amiodaron Hcl Hikma | Sedacoron;
- (AU) Australia: Amdarone | Amioðarone | Amioðarone actavis | Amioðarone sandoz | Apo Amioðarone | Aratac | Cardinorm | Cm amioðarone | Corðarone x | Rithmik | Tw amioðarone;
- (BD) Bangladesh: Cardiron | Pacet;
- (BE) Belgium: Amioðarone ab | Amioðarone bexal | Amioðarone Eurogenerics | Amioðarone merck-generics | Amioðarone sandoz | Corðarone;
- (BF) Burkina Faso: Corðarone | Eurythmic;
- (BG) Bulgaria: Amiocordin | Amiodaron | Amioðarone | Amioðarone Nihfi | Amiohexal | Corðarone;
- (BR) Brazil: Amiobal | Amiodarona | Amiodex | Amioron | Ancoron | Angiodarona | Angyton | Atlansil | Cardicoron | Cardiron | Cloridrato de amiodarona | Cor mio | Diodarone | Hiperterona | Miocor | Miocoron | Miodarid | Miodaron | Miodon | Taquicord;
- (CH) Switzerland: Amiodar | Amiodaron Labatec | Amiodaron-mepha | Amioðarone Zentiva | Corðarone | Escodaron;
- (CI) Côte d'Ivoire: Amioðarone aguettant | Eurythmic;
- (CL) Chile: Amiodarona | Atlansil | Corðarone | Ritmocardyl;
- (CN) China: Amioðarone | Corðarone | Ke da long | Sedacoron | Yi jie an;

- (CO) Colombia: Amiodarona | Amiodarona clorhidrato | Amiorit | Asulblan | Cirtrent | Coraaron | Daronal;
- (CR) Costa Rica: Amiodarona clorhidrato | Coraaron;
- (CZ) Czech Republic: Amiodaron hameln | Amiodaron Mylan | Amiohexal | Amiokordin | Coraaron | Rivodaron | Sedacoron;
- (DE) Germany: Amiobeta | Amiod | Amiodarex | Amiodaron | Amiodaron - 1 a pharma | Amiodaron aurobindo | Amiodaron beta | Amiodaron Carino | Amiodaron hameln | Amiodaron hcl stragen | Amiodaron Heumann | Amiodaron holsten | Amiodaron puren | Amiodaron sandoz | Amiodaron stada | Amiodaron winthrop | Amiodaron-ratiopharm | Amiodaronhydrochlorid HEXAL | Amiodura | Amiogamma | Amiohexal | Cordarex | Coraaron | Cornaron | Eucard | Sedacoron | Tachydaron;
- (DK) Denmark: Cordan;
- (DO) Dominican Republic: Amiocar | Amiodarona | Amiodarona MK | Amioran | Amiorit | Atlansil | Cardiogesic | Coraaron | Coronat | Isorrit | Miodafran | Miodar | Mionor;
- (EC) Ecuador: Acrodarona | Amiodarona | Amiodarona clorhidrato | Amiodarona MK | Amiodarox | Atlansil | Coraaron | Ritmocardyl;
- (EE) Estonia: Amioaaron-alpha | Amiokordin | Cordaron | Coraaron | Sedacoron;
- (EG) Egypt: Amiron | Cardio mep | Coraaron | Farcodaron | Roncard | Sedacoron | Ventromed;
- (ES) Spain: Amiodarona aurovitas | Trangorex;
- (ET) Ethiopia: Amioaaron sandoz;
- (FI) Finland: Amiodaron hameln | Coraaron;
- (FR) France: Amioaaron | Amioaaron actavis | Amioaaron aguettant | Amioaaron Almus | Amioaaron Alter | Amioaaron arrow | Amioaaron biogaran | Amioaaron cristers | Amioaaron g gam | Amioaaron gnr | Amioaaron irex | Amioaaron ivax | Amioaaron merck | Amioaaron Qualimed | Amioaaron ratiopharm | Amioaaron rpg | Amioaaron sandoz | Amioaaron teva | Amioaaron Zydu | Corbionax | Coraaron;
- (GB) United Kingdom: Amidox | Amioaaron | Amioaaron Almus | Amioaaron arrow | Amioaaron cox | Amioaaron focus | Amioaaron kent | Amioaaron Medreich | Amioaaron sandoz | Amyben | Coraaron x | Rotaritmil;
- (GR) Greece: Angoron;
- (HK) Hong Kong: Amioaaron HCL | Amioaaron sandoz | Apo Amioaaron | Coraaron | Seda-
coron;
- (HR) Croatia: Amiodaron | Amiokordin | Amlox | Coraaron;
- (HU) Hungary: Amiodarona lph | Amiohexal | Amiokordin | Coraaron | Sedacoron;
- (ID) Indonesia: Coraaron | Kendaron | Rexitron | Tiaryt;
- (IE) Ireland: Coraaron x;
- (IL) Israel: Amiodacore | Procor;
- (IN) India: Amidon | Amiodar | Amiodon | Amione | Amipace | Cardaron | Cardasol | Coraaron | Duron | Eurythmic | Pacenorm | Panaron | Ritebeat | Tachyra;
- (IQ) Iraq: Amicord;
- (IT) Italy: Amiodar | Amioaaron | Amioaaron aurobindo | Amioaaron cloridrato bioindustria | Amioaaron eg | Amioaaron ratiopharm | Coraaron;
- (JO) Jordan: Coraaron | Sedacoron;
- (JP) Japan: Amioaaron hydrochloride te | Amioaaron hydrochloride towa | Ancaron;
- (KE) Kenya: Coraaron | Eurythmic;
- (KR) Korea, Republic of: Coraaron;
- (KW) Kuwait: Coraaron;
- (LB) Lebanon: Coraaron | Rivodaron | Sedacoron;
- (LT) Lithuania: Amiodaron | Amiodarona aurobindo | Amioaaron | Amioaaron actiopharma | Amiokordin | Coraaron | Sedacoron;
- (LU) Luxembourg: Amioaaron Eurogenerics | Coraaron;
- (LV) Latvia: Amiocordin | Amioaaron | Coraaron;
- (MA) Morocco: Coraaron;
- (MX) Mexico: Amiodarona | Braxan | Cirtrent | Coraaron | Darfin | Defaven | Keritmon | Kuncar | Ranfodar | Tecnovid;
- (MY) Malaysia: Amioaaron hcl bioindustria | Amiohexal | Aratac | Cardilor | Coraaron | Euryth-
mic;

- (NL) Netherlands: Amiodaron | Amiodaron gf | Amiodaron Hcl | Amiodaron hcl Aurobindo | Amiodaron Hcl Hikma | Amiodaron hcl sandoz | Amiodaron HCl Teva | Amiodaron kwk | Corðarone;
- (NO) Norway: Amiodaron | Amiodaron Hcl Hikma | Amioðarone | Amioðarone gevita | Corðarone;
- (NZ) New Zealand: Amioðarone HCL | Amioðarone hydrochloride hameln | Aratac | Corðarone x;
- (PA) Panama: Amiodarona | Amiodarona hameln | Corðarone | Miodarona;
- (PE) Peru: Amiodarona | Atlansil | Corðarone | Ritmax | Ritmocardyl;
- (PH) Philippines: Amioðarone sandoz | Amiron | Anoion | Corðarone | Daryth | Eurythmic | Medoðarone | Myodial | Rythma | Triodone;
- (PK) Pakistan: Aldaron | Corðarone x | Sedacoron;
- (PL) Poland: Amiodaron hameln | Amiokordin | Corðarone | Opacorden | Sedacoron;
- (PR) Puerto Rico: Amioðarone | Amioðarone HCL | Corðarone | Nexterone | Pacerone;
- (PT) Portugal: Amiodarona | Amiodarona aurobindo | Corðarone | Miodrone;
- (PY) Paraguay: Acetal | Amiodar | Amiodarona clorhidrato | Amiodarona la sante | Atlansil | Corðarone | Miodaron | Prinox;
- (QA) Qatar: Amiodaron | Cordalin | Corðarone | Sedacoron;
- (RO) Romania: Amiodarona | Amiodarona arena | Amiodarona lph | Amiokordin | Corðarone | Daritmin;
- (RU) Russian Federation: Amiocordin | Amiodaron | Amiodaron FPO | Amiodaron hexal | Amioðarone | Amioðarone sandoz | Amiokordin | Cardiodaron | Corðarone | Rhythmiðarone | Sedacoron | Vero amioðarone;
- (SA) Saudi Arabia: Amirone | Corðarone | Pms-amioðarone | Sedacoron;
- (SE) Sweden: Amiodaron Evolan | Amiodaron hameln | Amiodaron stragen | Corðarone;
- (SG) Singapore: Amioðarone ph&t | Aratac | Corðarone;
- (SI) Slovenia: Amiokordin | Amjodaron Hameln | Corðarone;
- (SK) Slovakia: Amiodaron | Amiohexal | Amiokordin | Corðarone | Rivodaron | Sedacoron;
- (SR) Suriname: Amiodaron | Amiodaron Hcl | Amiodaron hcl apotex | Amiodaron hcl mylan | Amioðarone | Amioðarone ratiopharm | Cardilor;
- (SV) El Salvador: Amiodarona | Cirtrent;
- (TH) Thailand: Aldarone | Amdarone | Amidarone | Amioðarone aguettant | Aratac | Cardilor | Corðarone | Eurythmic | Zydaron;
- (TN) Tunisia: Amiocard | Amioðarone | Amioðarone aguettant | Corðarone | Ricardyl | Rythmarone | Sedacoron;
- (TR) Turkey: Corðarone;
- (TW) Taiwan: Adarone | Amiorone | Corðarone | Harten | Rythmarone | Sedacoron;
- (UA) Ukraine: Amiocordin | Amiodaron | Amiodaron darnitsa | Aritmil | Cardiodaron | Corðarone | Miorytmil | Rotaritmil | Sedacoron;
- (UG) Uganda: Corðarone;
- (UY) Uruguay: Amiodarona | Amiodarona icu vita | Amiodarona lazar | Atlansil | Miodorit | Taquirona | Trangorex;
- (VE) Venezuela, Bolivarian Republic of: Amiodarona | Amiodarona clorhidrato | Amioran | Arycor | Cardiogesic | Coradona | Diarona | Novarona | Prismadarone | Trangorex;
- (ZA) South Africa: Activo amioðarone | Adcorone | Amiotach | Arycor | Bio Amioðarone | Corðarone x | Hexarone

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